0DTE Trading Rules

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Abstract

We study realized payoffs of S&P500 zero-days-to-expiration (0DTE) options and standard multi-leg structures from 09/2016 to February 2, 2026. A positive 0DTE variance risk premium exists, but at same-day horizons its economic magnitude is small and difficult to monetize after realistic frictions. Across individual options and strategy templates, net PNL distributions are wide, state-dependent, and dominated by tail risk rather than by stable mean carry. Ex-post strategy PNL loads more on directional and skewness realizations than on realized variance alone. At the same time, disciplined 10:00 ET conditional rules estimated under a strict out-of-sample protocol deliver economically meaningful net performance for selected strategies and diversified baskets. The evidence, therefore, points to selective timing opportunities rather than to a broad unconditional edge.

Keywords: 0DTE, ultra-short options, variance risk premium, volatility trading, option strategies, option trading

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1 Introduction

Trading in zero-days-to-expiration options has become one of the most visible recent developments in the U.S. index-option market. This growth raises two natural questions. The first is descriptive: what do implementable 0DTE S&P500 (option root SPXW) positions actually earn once payoffs are marked consistently, scaled comparably across strikes, and adjusted for execution frictions? The second is conditional: if these payoffs are strongly state-dependent, can that variation be converted into real-time trading rules rather than into ex-post stories?

We study these questions using SPXW 0DTE calls, puts, and standard multi-leg structures over 09/2016 to February 2, 2026. The main text focuses on positions opened at 10:00 ET and held to the 16:00 ET close; the Appendix reports analogous evidence for entries at 16:00 ET on the previous day and at 13:00 ET and 15:00 ET on the same day. At each entry time, we interpolate the observed 0DTE cross-section over moneyness from 0.98 to 1.02 in steps of 0.001 and use these standardized instruments to construct comparable option- and strategy-level payoffs. The strategy set covers straddles and strangles, iron butterflies and iron condors, bull call and bear put spreads, call and put ratio spreads, and risk reversals.

Three findings stand out. First, a positive 0DTE variance risk premium exists, but at same-day horizons its economic magnitude is small. Second, strategy-level payoff distributions are wide, tail-heavy, and unstable across regimes; for many structures, downside risk is large relative to average carry. Third, conditional timing works better when formulated as directional classification than when posed as direct return prediction. In the conditional out-of-sample (OOS) implementation, put ratio spreads reach a net Sharpe ratio of 1.26, iron butterfly structures reach 0.82, and diversified equal-weight baskets reach net Sharpe ratios between 1.01 and 1.27.

We contribute to several adjacent strands of research. Relative to the emerging 0DTE market-impact literature, including the merged draft of Adams, Dim, Eraker, Fontaine, Orn-

thanalai, and Vilkov 2025[1] and the related evidence in Brogaard, Han, and Won 2026, Amaya, Garcia-Ares, Pearson, and Vasquez 2025, our object is not the causal effect of dealer hedging on aggregate volatility, but the realized payoff distribution of standardized strategy templates under implementable construction rules. Relative to the ultra-short option-pricing literature (e.g., Bandi, Fusari, and Renò 2023, Almeida, Freire, and Hizmeri 2025), we take a reduced-form route and document how feasible 0DTE strategy PNL maps into implied and realized moments. Relative to the broader short-dated option-return literature (e.g., Beckmeyer, Branger, and Gayda 2023, Bryzgalova, Pavlova, and Sikorskaya 2023, Bogousslavsky and Muravyev 2025), our evidence supports the view that gross option stories weaken materially once one accounts for execution costs and tail risk. Finally, relative to the option-return and event-state literature (e.g., Büchner and Kelly 2022, Londono and Samadi 2023, Knox, Londono, Samadi, and Vissing-Jorgensen 2025), we study unhedged same-day strategy payoffs, restrict predictors to information available by 10:00 ET, and impose a strict out-of-sample protocol for conditional rules.

The rest of the paper proceeds as follows. Section 2 describes the data and the construction of the option-, strategy-, and state variables used throughout. Section 3 documents unconditional payoffs of individual options and strategy templates. Section 4 links strategy PNL to implied and realized moments and tightens inference for these links. Section 5 studies conditional trading rules. Section 6 concludes. Appendix A contains supporting tables and figures.

2 Data and Variable Construction

Data on Options and Underlying Markets. This section describes the data sources and the construction choices used throughout the paper. We use Cboe 30-minute option bars with

[1]The 2025 merged draft consolidates and extends two predecessor papers, Adams, Fontaine, and Ornthanalai 2025 and Dim, Eraker, and Vilkov 2024.

NBBO quotes and sizes, OHLC prices, trade volume, and underlying levels, focusing on SPX Weeklys (root `SPXW`). These contracts are European and cash settled against the SPX close at 16:00 ET. The sample runs from 09/2016 to February 2, 2026. SPXW had three weekly expirations from late August 2016; two additional weekdays were introduced on April 18, 2022 and May 11, 2022, giving daily weekday 0DTE expirations. For underlying prices and realized-moment construction, we use *ThetaData* one-minute bars (SPX and VIX where needed) and ThetaData daily closes.

Implied and Realized Moment Measures. We compute implied variance (IV) to expiration at each intraday bar-end t on expiration day using the VIX methodology Cboe (2023) applied to SPXW 0DTE options. The conditioning variable at 10:00 ET is $IV_{10:00}$, i.e., integrated implied variance from 10:00 ET to 16:00 ET:

$$IV_t = 2e^{rT} \sum_i \frac{\Delta K_i}{K_i^2} Q(K_i) - [F/K_0 - 1]^2, \tag{1}$$

where K_i is strike, K_0 is the first strike at or immediately below option-implied forward F, $Q(K_i)$ is the OTM mid quote ($i \neq 0$), $Q(K_0)$ is the average of call and put quotes at K_0, r is the risk-free rate (1-month Treasury bill from FRED), and T is time to expiration in years. We also define up and down semivariances, IV_t^{up} and IV_t^{dn}, by applying (2) to weakly OTM calls and puts, respectively, and scaling the subtracted term to $0.5[F/K_0 - 1]^2$ so that the two components sum to total IV_t. The difference between up and down semivariance is a common implied-skewness proxy (e.g., Feunou, Jahan-Parvar, and Okou (2018), Kilic and Shaliastovich (2019)), and we use: [2]

$$IS_t = IV_t^{up} - IV_t^{dn}. \tag{2}$$

[2]Implied skewness can also be computed directly (e.g., Bakshi, Kapadia, and Madan (2003), Kozhan, Neuberger, and Schneider (2013)). We use the semivariance-difference proxy because it is simple under both risk-neutral and physical measures and maps naturally to skewness-premium definitions.

We compute forward-looking realized variance (RV) at the end of each bar t for periods matching each computed IV as the sum of squared one-minute log returns from the end of a bar to the end of the day:

$$RV_t = \sum_{t}^{T-1} r_{t,t+1}^2, \tag{3}$$

where $r_{t,t+1}$ is the one-minute close-to-close log return from minute t to $t+1$, summed through T=16:00 on the same day. Analogous to implied semivariances, realized semivariances RV_t^{up} and RV_t^{dn} are computed by multiplying each $r_{t,t+1}^2$ term in (3) by $\mathbf{1}_{r_{t,t+1}>0}$ or $\mathbf{1}_{r_{t,t+1}<0}$, respectively. We define realized skewness proxy as

$$RS_t = RV_t^{up} - RV_t^{dn}. \tag{4}$$

We define *ex post* variance risk premium VRP at bar t as the payoff on a short variance swap from t to expiration, i.e., implied minus realized variance over that window:

$$VRP_t = IV_t - RV_t. \tag{5}$$

Semivariance premia VRP_t^{up} and VRP_t^{dn} are defined analogously.[3] Skewness premium is then defined in two equivalent ways:

$$SRP_t = IS_t - RS_t = VRP_t^{up} - VRP_t^{dn}. \tag{6}$$

For options expiring the same day, observed at an intraday time (for example 10:00 ET), we compute moneyness as strike divided by spot and retain options in $[0.98, 1.02]$ (within $\pm 2\%$ of ATM). We scale mid prices and bid-ask spreads by spot, then interpolate key variables (mid, spread, implied volatility, Greeks, and leg-level flow/liquidity inputs used later) over moneyness

[3] We do not annualize these quantities; the goal is to keep magnitudes in realized same-day payoff units.

with step 0.001, separately for calls and puts.[4] For each interpolated option, we compute the payoff:

$$Payoff_t(M, Type) = \begin{cases} (S_T/S_t - M)^+ & \text{if } Type = Call, \\ (M - S_T/S_t)^+ & \text{if } Type = Put, \end{cases} \tag{7}$$

where M is moneyness and S_T/S_t is gross index return from t to T=16:00. For each option we also compute intrinsic value (immediate-exercise payoff) and time value (mid minus intrinsic). If time value is negative because of mid-quote noise, we set time value to zero and intrinsic value to mid. Since moneyness is strike scaled by spot, payoff, intrinsic, and time value are already in spot-relative units. Option PNL relative to spot is:

$$PNL_t(M, Type) = Payoff_t(M, Type) - Mid_t(M, Type), \tag{8}$$

where $Mid_t(M, Type)$ is the mid price at time t for option $(M, Type)$. Realized return to expiration is:

$$Ret_t(M, Type) = \frac{PNL_t(M, Type)}{Mid_t(M, Type)} = \frac{Payoff_t(M, Type)}{Mid_t(M, Type)} - 1. \tag{9}$$

We report both PNL and return in percentage terms (multiply by 100). For multi-leg structures, strategy PNL is the signed sum of leg-level PNL weighted by leg quantities; by construction it remains in spot-relative units.

Construction of Conditional Features at 10:00 ET. For the conditional model section, we use the same SPXW option pipeline and then construct additional predictors at the strategy level. In preprocessing, we keep same-day-expiry SPXW quotes for intraday bars and include

[4] At SPX= 4000, step 0.001 corresponds to roughly 4 index points. Interpolation standardizes moneyness across dates and enables direct strategy construction. We use one-dimensional piecewise cubic interpolation (Akima 1970), with similar results under alternative interpolation schemes. Interpolation is cross-sectional in moneyness at a fixed date/time and option type; we do not interpolate across time.

next-day-expiry contracts only at 16:00 ET for close-to-close alignment in intermediate steps; the conditional models themselves use the 10:00 ET cross-section of same-day-expiry strategy observations only. Calls and puts are processed separately: interpolation is run within option type on the moneyness grid $M \in \{0.98, 0.981, \ldots, 1.02\}$, and strategy legs are matched by $(datetime, option_type, moneyness)$. In short, option-leg predictors are taken from this interpolated 10:00 ET grid, while market controls and lagged realized variables are taken from non-interpolated underlying time series.

Let $i = (s, m, t)$ index strategy type s, moneyness configuration m, and date-time t (here, $t =$ 10:00 ET), and let $\mathcal{L}_{s,m}$ denote legs in that structure, with signed leg quantity q_l ($q_l > 0$ long, $q_l < 0$ short). For each leg l, we collect the option inputs bas_l, mid_l, Δ_l, Γ_l, ν_l, Vol_l, OI_l, $BidSz_l$, $AskSz_l$, and the active underlying level S_l. If multiple rows map to the same $(datetime, option_type, moneyness)$ key, we aggregate by variable type before leg construction: price and Greek variables $(bas, mid, \Delta, \Gamma, \nu, S, OI)$ are averaged, while traded and displayed depth variables $(Vol, BidSz, AskSz)$ are summed. For compactness, we denote half-spread by h, displayed depth by d, relative spread by ρ, traded volume by v, flow notional by f, and gamma exposure by g.

We then define leg-level flow, gamma-exposure, and liquidity building blocks. In our notation, (h, d, ρ) are *liquidity* features. They proxy execution cost, displayed depth, and relative tightness of quoted markets at the entry time:

$$h_{i,l} = |q_l| \cdot \frac{bas_l}{2}, \tag{10}$$

$$d_{i,l} = |q_l| \cdot (BidSz_l + AskSz_l), \tag{11}$$

$$\rho_{i,l} = |q_l| \cdot \frac{bas_l}{|mid_l|}. \tag{12}$$

The variables $(v, f^{\Delta}, f^{\Gamma}, f^{\nu})$ are *flow* features: they measure how much contract volume, delta notional, gamma notional, and vega notional trade through the strikes used by the strategy:

$$v_{i,l} = |q_l| \cdot Vol_l, \tag{13}$$

$$f^{\Delta}_{i,l} = |q_l| \cdot Vol_l \cdot |\Delta_l| \cdot 100 \cdot S_l, \tag{14}$$

$$f^{\Gamma}_{i,l} = |q_l| \cdot Vol_l \cdot |\Gamma_l| \cdot 100 \cdot S_l^2, \tag{15}$$

$$f^{\nu}_{i,l} = |q_l| \cdot Vol_l \cdot |\nu_l| \cdot 100. \tag{16}$$

The variables $(g^{OI,n}, g^{OI,a}, g^{\Gamma,n}, g^{\Gamma,a})$ are *GEX-style* features. They summarize signed and absolute convexity exposure, either weighted by open interest or coming directly from the strategy-leg mix:

$$g^{OI,n}_{i,l} = q_l \cdot OI_l \cdot \Gamma_l \cdot 100 \cdot S_l^2, \tag{17}$$

$$g^{OI,a}_{i,l} = |q_l| \cdot OI_l \cdot |\Gamma_l| \cdot 100 \cdot S_l^2, \tag{18}$$

$$g^{\Gamma,n}_{i,l} = q_l \cdot \Gamma_l, \tag{19}$$

$$g^{\Gamma,a}_{i,l} = |q_l| \cdot |\Gamma_l|. \tag{20}$$

We derive these three families separately because they capture distinct economic channels: implementation frictions, current trading pressure, and concentration of convexity.

The leg-level liquidity, flow, and GEX features defined above are aggregated to the strategy level by summing over all legs in the strategy at 10:00 ET. In the current conditional models we use same-day-expiry contracts only at 10:00 ET; next-day-expiry contracts are used only for

16:00 ET alignment in preprocessing. We additionally use transformed ratios:

$$B_i^{\Gamma} = \frac{g_i^{OI,n}}{|g_i^{OI,a}| + 1}, \tag{21}$$

$$R_i^{\Gamma} = \frac{f_i^{\Gamma}}{|g_i^{OI,a}| + 1}, \tag{22}$$

$$T_i = \frac{h_i}{d_i + 1}. \tag{23}$$

These transformed indicators remain in the same three families: B^{Γ} is a normalized *GEX balance* measure, R^{Γ} is a *flow-pressure* measure scaled by the local GEX base, and T is a *liquidity tightness* measure. We derive them because raw flow and raw GEX are strongly scale-dependent across strategies, strike widths, and premium sizes. These are *flow-style* and exposure-style proxies based on traded volume, open interest, and leg Greeks; they are not a dealer-inventory reconstruction. If a strategy observation has incomplete leg coverage at 10:00 ET, we set the engineered leg-aggregated features to missing. For specifications that use date-wise cross-sectional scaling, each strategy-level predictor $X_{i,t}$ is transformed as

$$X_{i,t}^{CS} = \frac{X_{i,t} - \overline{X}_t}{std_t(X)}, \quad \overline{X}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} X_{i,t}, \tag{24}$$

where N_t is the number of strategy observations at date t.

Baseline strategy-level predictors include time value, mid-price, Δ, Γ, ν, moneyness-center, and lagged PNL terms. For a structure with leg moneyness values $\{M_l\}_{l \in \mathcal{L}_{s,m}}$, we define

$$M_{i,t} = \frac{1}{|\mathcal{L}_{s,m}|} \sum_{l \in \mathcal{L}_{s,m}} M_l. \tag{25}$$

Lagged PNL features are

$$P_{i,t}^{(1)} = PNL_{i,t-1}, \quad \bar{P}_{i,t}^{(5)} = \frac{1}{5} \sum_{j=1}^{5} PNL_{i,t-j}, \quad \sigma_{P,i,t}^{(5)} = sd(PNL_{i,t-1}, \ldots, PNL_{i,t-5}), \tag{26}$$

computed within each $(strategy, moneyness)$ series. We also include strategy fixed identifiers as one-hot dummies $D_{s,i} \in \{0,1\}$. Market-state controls at 10:00 ET are

$$IV_t = 10^5 \cdot \overline{VIX}_t^{(SPXW,0DTE,10:00)}, \tag{27}$$

$$IS_t = 10^5 \cdot \left(\overline{VIX^{up}}_t^{(SPXW,0DTE,10:00)} - \overline{VIX^{dn}}_t^{(SPXW,0DTE,10:00)} \right), \tag{28}$$

plus slope proxies $(slope_up_t, slope_dn_t)$ and one-day-lagged realized SPX moments, namely lagged realized returns, variance, and skewness, where the lag enforces strict no-look-ahead construction. These variables form the remaining two families. The *baseline* family consists of the strategy moneyness center M, yesterday's strategy PNL $P^{(1)}$, the five-day average strategy PNL $\bar{P}^{(5)}$, the five-day standard deviation of strategy PNL $\sigma_P^{(5)}$, and the option-level descriptors used elsewhere in the models (time value, mid price, Δ, Γ, and ν); together they capture strategy geometry, local carry, and short-horizon persistence. The *market-state* family consists of (IV, IS), slope proxies, and lagged realized SPX measures; they capture the entry-time variance/skew environment and recent underlying conditions observed by 10:00 ET. We derive these families to separate strategy-specific signals from market-wide states. The continuous target in the model zoo is the strategy return-per-underlying $y_{i,t} = reth_und_{i,t}$, and the directional target is $\mathbb{1}(y_{i,t} > 0)$. A full predictor dictionary for the compact notation used here is reported in Appendix Table A3.

3 Unconditional Trading Rules

An unconditional strategy opens option positions each day without conditioning on state variables that would change timing or size. Positions are then held to expiration, and performance is measured by holding-period return and PNL. The simplest positions use a single option; richer structures combine multiple legs to target specific payoff and Greeks profiles.

We compute return and PNL to expiration for all individual options and strategy combinations and summarize their distributions.

3.1 Variance Risk Premium

Over the full sample (09/2016 to February 2, 2026), 0DTE options deliver a positive and statistically significant variance risk premium: implied variance exceeds realized variance to expiration (Figure 1, Panel B). Trading variance requires a strip of options across strikes, with weights

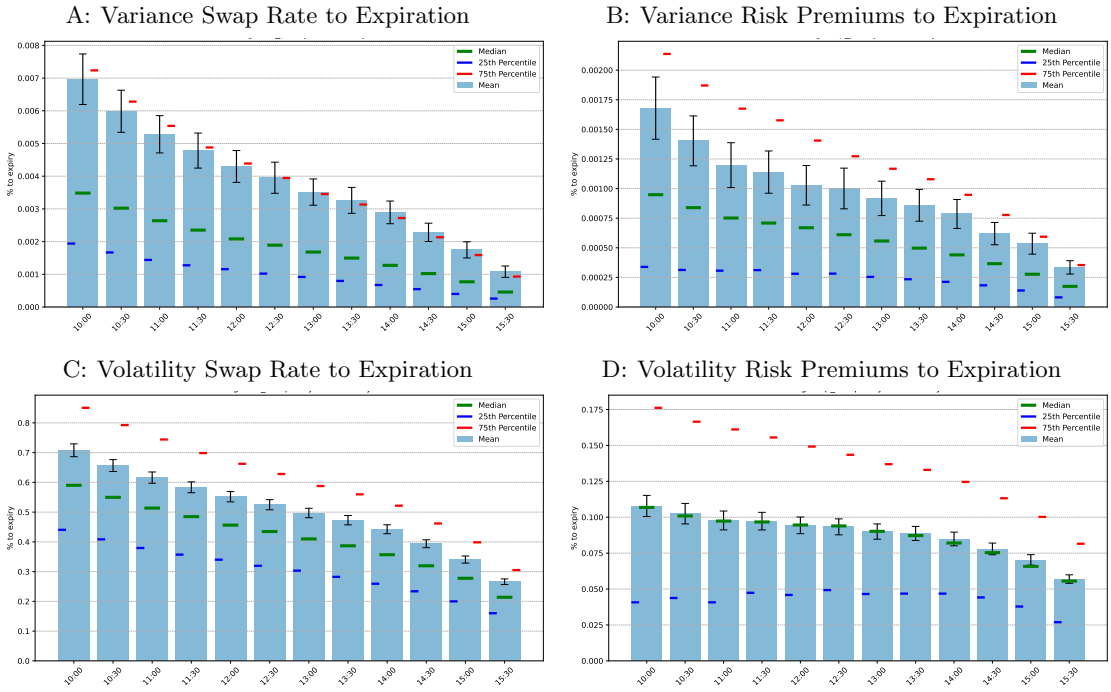


Figure 1: Variance Risk Premiums. The figure shows average variance and its risk premiums (VRP) to expiration for 0DTE SPXW options by intraday 30-minute bars. VRP is computed as implied minus realized variances from a given bar to expiration at 16:00 ET. Panels C and D are based on average volatility and differences in volatilities, respectively. We average variables measured at the end of each bar to expiration at 16:00 that day (with 95% confidence bounds based on Newey-West standard errors with three lags; Newey and West 1987). Bars show mean values; each bar is accompanied by median, 25th, and 75th percentiles. X-axis labels show the endpoints of intraday bars. The sample period is from 09/2016 to February 2, 2026.

determined by the chosen model-free variance formula. Under the Cboe VIX construction, risk-neutral variance is a weighted sum of OTM option prices (equation 2). In practice, the 0DTE variance swap rate to expiration is a weighted sum of OTM call and put time values (Figure 1, Panel A). The realized variance risk premium can be interpreted as time value retained after

paying exercised options at expiration. As realized variance increases, terminal index moves are larger on average and payouts rise; with zero realized variance, the index is unchanged and the initial time value is retained.

Panel B of Figure 1 suggests that selling 0DTE variance is profitable on average. The economic magnitude, however, is small because OTM time values only hours before expiration are small (Figure 2). Even in an extreme zero-realized-variance scenario, collecting roughly twice the average OTM time value implies only about 0.20% of spot. Translating the variance object into a spot-relative monetization benchmark, the median realized VRP from 10:00 ET to expiration is approximately 0.0011% of underlying. These magnitudes are difficult to monetize after realistic trading frictions.

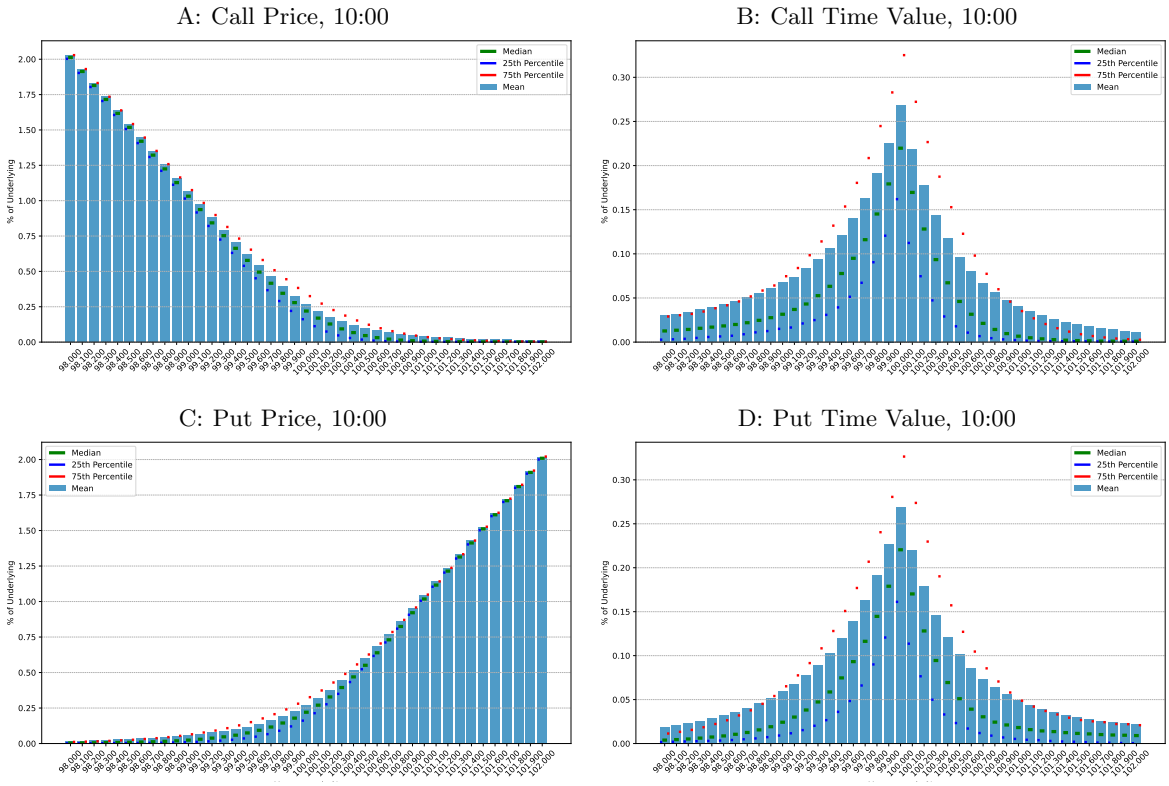


Figure 2: 0DTE Option Prices and Time Value. The figure provides statistics on prices of 0DTE call and put options at 10:00 ET. Panels on the left show option mid-price relative to the underlying price in %. Panels on the right show time value relative to underlying, also in %. Bars show mean values, and each bar is accompanied by median, 25th, and 75th percentiles. X-axis labels show the moneyness of the analyzed options. The sample period is from 09/2016 to February 2, 2026.

3.2 Individual Options

For individual calls and puts, we report both return relative to mid price and realized PNL relative to spot at entry, $(payoff - mid\ price)/underlying\ price \times 100\%$, using 10:00 ET entries in the main text (Figure 3). Individual option returns are highly volatile, especially for OTM contracts, and mean returns are often not statistically distinct from zero. Because return distributions are strongly skewed, inference on means alone is not very informative. PNL scaled by spot is less asymmetric and shows that ATM-to-OTM calls and most puts lose on average, while OTM contracts have relatively tighter distributions. Selling slightly OTM calls and puts is positive in up to 75% of observations; deeper ITM contracts are much more volatile, with interquartile ranges around 0.7–0.8% of spot.

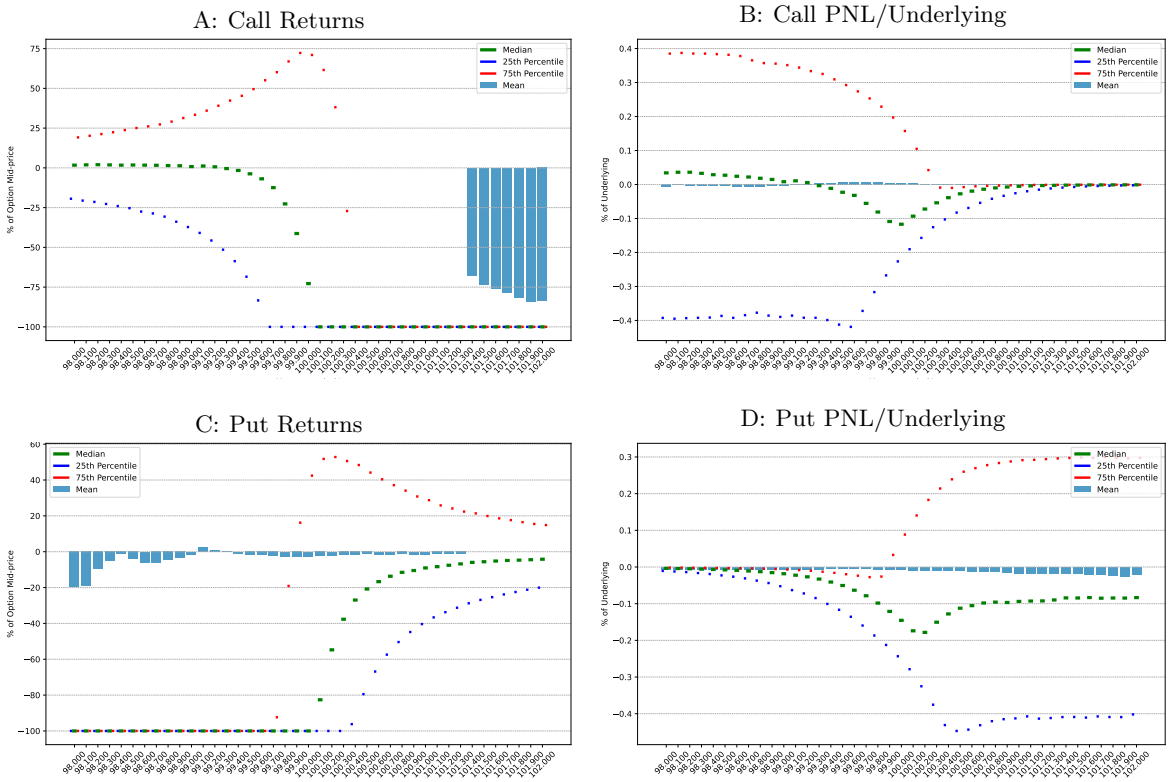


Figure 3: 0DTE Option Returns. The figure provides statistics on the profitability of naked 0DTE call and put option buying at 10:00 ET and holding to expiry at 16:00 ET. Panels on the left show realized return in % relative to option mid-price. Panels on the right show the realized profit per one unit of underlying relative to underlying price $(payoff - mid\ price)/underlying\ price \times 100\%$. Bars show mean values, and each bar is accompanied by median, 25th, and 75th percentiles. X-axis labels show the moneyness of the analyzed options. The sample period is from 09/2016 to February 2, 2026.

Option performance for entries at 16:00 ET on the previous day, 13:00 ET, and 15:00 ET is shown in Figures A2, A3, and A4. The main qualitative findings are unchanged across entry times.

3.3 Option Strategies

Single-option positions load on both volatility and direction, while multi-leg structures allow tighter exposure design. We analyze: (i) ATM straddles and strangles, which are close to delta-neutral at initiation and primarily volatility-focused; (ii) iron butterflies and iron condors, which are short-volatility with bounded downside; (iii) risk reversals, which load on skewness premia; (iv) bull call and bear put spreads, which are directional structures; and (v) call and put ratio spreads (1×2), which target moderate directional moves with asymmetric tails.

These structures have distinct payoff profiles (Figure A1). Straddles/strangles have limited downside and open upside; iron butterfly/condor have bounded upside and downside; bull/bear spreads are directional with bounded payoffs; ratio spreads have capped upside and open downside; risk reversals are open-tailed on both sides. Entry cost reflects these shapes: narrow-strike risk reversals and ratio spreads are often credit structures, while straddles and strangles are typically the most expensive debit structures. For some combinations, the initial net premium is close to zero or negative, so the return scaled by the initial premium is unstable or not economically meaningful. We therefore focus on realized PNL scaled by spot. Figure 4 and Table 1 show broad dispersion for most structures. To keep the table readable, we report only near-ATM combinations there, while Figure 4 continues to show the full grid. Spread and ratio-spread outcomes are highly variable, with interquartile ranges often roughly symmetric around zero. For ATM-long directional spreads, median PNL is typically negative. Using ITM long legs yields more symmetric distributions with means close to zero and small sign differences between call and put structures, consistent with the upward index drift in much of 2017–2023. OTM

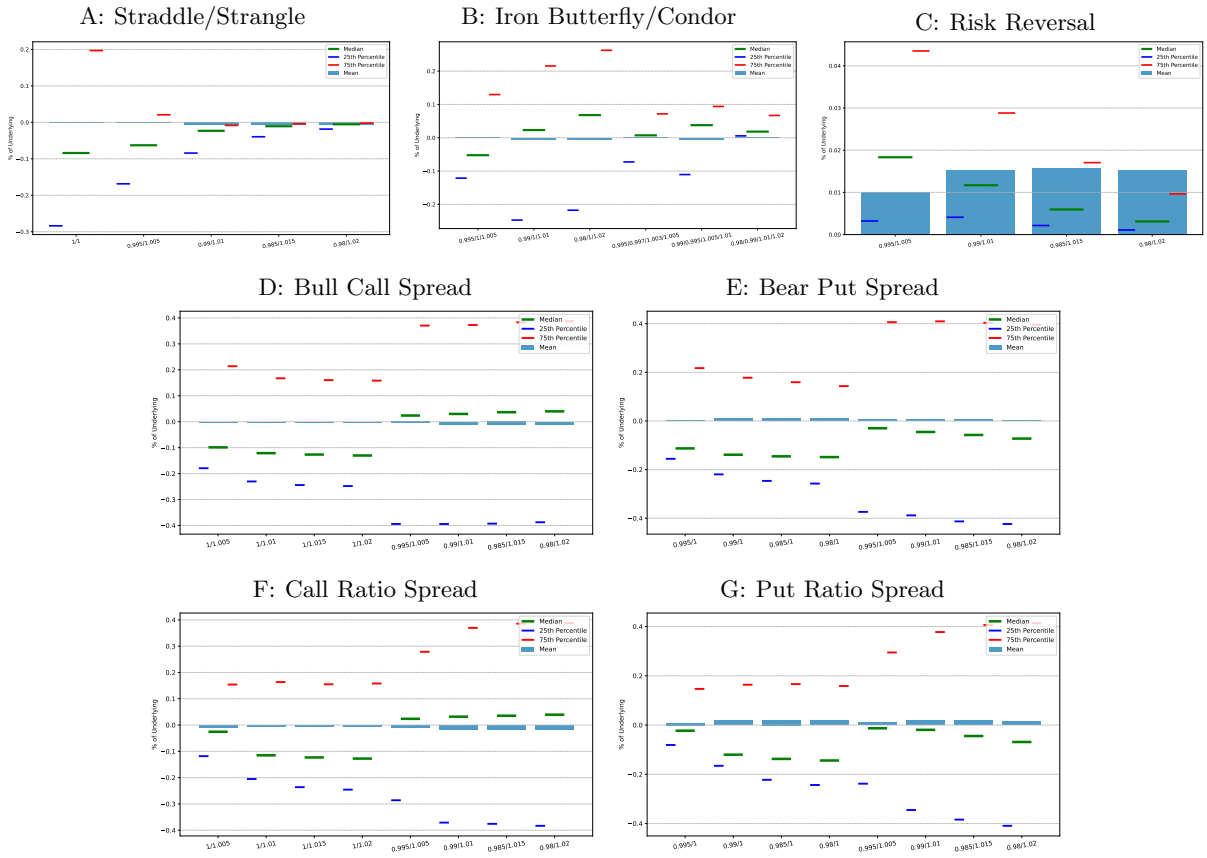


Figure 4: 0DTE Static Option Strategies. The figure shows statistics on the profitability of 0DTE option strategies from 10:00 ET to expiry at 16:00 ET. All panels show strategies' realized PNL relative to underlying price $(payoff - mid\ price)/underlying\ price \times 100\%$. Bars show mean values, accompanied by the median and the 25th and 75th percentiles. X-axis labels show the combination of moneyness of options used for a strategy. The sample period is from 09/2016 to February 2, 2026.

strangles are relatively compact and mostly negative in the upper and lower quartiles, so short strangles outperform in at least 75% of observations. Risk reversals are the only structure with consistently positive mean, median, and 25th percentile PNL, but the average effect is small (about 0.01% of spot), which limits implementability.

Strategy	Moneyness	Count	Mean	Volatility	Min	1%	25%	50%	75%	99%	Max	Skew	SR, p.a.
Strangle/Straddle	1/1	1318	-0.0011	0.55	-2.69	-0.99	-0.28	-0.08	0.20	1.83	6.04	2.11	-0.03
	0.995/1.005	1319	-0.0009	0.47	-2.70	-0.83	-0.17	-0.06	0.02	1.76	6.02	3.07	-0.03
	0.99/1.01	1319	-0.0067	0.38	-2.75	-0.78	-0.08	-0.02	-0.01	1.59	5.93	4.69	-0.28
Iron Butterfly/Condor	0.995/1/1.005	1318	0.0002	0.16	-0.50	-0.25	-0.12	-0.05	0.13	0.35	0.43	0.48	0.02
	0.99/1/1.01	1318	-0.0056	0.30	-1.00	-0.61	-0.25	0.02	0.22	0.59	0.83	-0.14	-0.30
	0.995/0.997/1.003/1.005	1319	-0.0006	0.08	-0.20	-0.15	-0.07	0.01	0.07	0.15	0.18	0.00	-0.12
	0.99/0.995/1.005/1.01	1319	-0.0058	0.17	-0.50	-0.43	-0.11	0.04	0.09	0.29	0.40	-0.71	-0.54
Risk Reversal	0.995/1.005	1319	0.0100	0.62	-4.43	-1.86	0.00	0.02	0.04	1.94	8.28	2.04	0.25
	0.99/1.01	1319	0.0153	0.46	-3.89	-1.34	0.00	0.01	0.03	1.45	7.80	4.39	0.53
Bull Call Spread	1/1.005	1319	-0.0031	0.20	-0.43	-0.24	-0.18	-0.10	0.21	0.35	0.50	0.50	-0.24
	1/1.01	1319	-0.0028	0.32	-0.51	-0.42	-0.23	-0.12	0.17	0.74	1.00	0.92	-0.14
	0.995/1.005	1319	-0.0052	0.38	-0.70	-0.56	-0.39	0.02	0.37	0.50	1.00	-0.12	-0.22
	0.99/1.01	1319	-0.0113	0.58	-1.12	-1.07	-0.39	0.03	0.37	1.00	2.00	-0.14	-0.31
Call Ratio Spread	1/1.005	1319	-0.0076	0.37	-6.89	-1.10	-0.12	-0.03	0.15	0.52	1.80	-7.30	-0.33
	1/1.01	1319	-0.0071	0.37	-6.33	-0.45	-0.21	-0.12	0.16	0.73	1.26	-4.48	-0.31
	0.995/1.005	1319	-0.0097	0.43	-6.67	-0.89	-0.29	0.02	0.28	0.70	1.52	-3.84	-0.36
	0.99/1.01	1319	-0.0156	0.57	-5.91	-1.03	-0.37	0.03	0.37	1.03	1.64	-1.08	-0.43
Bear Put Spread	0.995/1	1318	0.0028	0.20	-0.27	-0.21	-0.16	-0.11	0.22	0.38	0.45	0.75	0.21
	0.99/1	1318	0.0084	0.33	-0.46	-0.38	-0.22	-0.14	0.18	0.81	0.89	1.12	0.40
	0.995/1.005	1269	0.0055	0.38	-0.58	-0.50	-0.37	-0.03	0.41	0.56	0.77	0.14	0.23
	0.99/1.01	1142	0.0064	0.59	-1.03	-1.00	-0.39	-0.05	0.41	1.07	1.12	0.18	0.17
Put Ratio Spread	0.995/1	1318	0.0082	0.32	-3.54	-1.30	-0.08	-0.02	0.15	0.59	1.43	-3.17	0.41
	0.99/1	1318	0.0194	0.32	-2.85	-0.53	-0.17	-0.12	0.16	0.83	1.99	0.23	0.95
	0.995/1.005	1269	0.0104	0.38	-3.29	-1.09	-0.24	-0.01	0.29	0.80	1.68	-1.00	0.43
	0.99/1.01	1142	0.0182	0.55	-2.40	-0.96	-0.35	-0.02	0.38	1.20	2.46	0.22	0.53

Table 1: 0DTE Static Option Strategies Performance. The table shows the summary statistics for the holding period PNL of 0DTE option strategies (from 10:00 ET to 16:00 ET) relative to underlying price $(payoff - mid\ price)/underlying\ price \times 100\%$. To keep the table concise, we report only near-ATM combinations with all legs within 1% of ATM; Figure 4 shows the full grid. The SR, p.a. is the Sharpe Ratio annualized by scaling it up by $\sqrt{252}$. The sample period is from 09/2016 to February 2, 2026.

Performance for the same strategy set at alternative entry times (16:00 ET previous day, 13:00 ET, and 15:00 ET) is shown in Figures A5, A6, and A7. Full-sample unconditional patterns are broadly similar across entry times. Subperiod tables for 2022–2023 and 01/2024–February 2, 2026(Tables A1 and A2) show substantial time variation. This motivates conditional rules that explicitly tie strategy exposure to observable market states.

3.4 Regime Stability Around the 2022 Expansion of Daily Expirations

To formalize the regime-shift discussion, we test whether average strategy PNL at 10:00 ET changed after the expansion to daily SPXW expirations in 2022. We define the pre-period as

dates up to April 14, 2022, the post-period as dates from May 11, 2022 onward, and exclude the transition window between April 18 and May 10, 2022. For each strategy, we estimate a pooled model with combo fixed effects and a post-2022 indicator, clustering standard errors by date. The post-2022 coefficient is the structural-break estimate.

Strategy	Pre-2022 Mean	Post-2022 Mean	Δ Post-Pre	$t(\Delta)$	p-value	Vol Ratio	N Pre	N Post
Strangle/Straddle	-0.0085	-0.0054	0.0031	0.15	0.877	1.04	2952	3575
Iron Butterfly/Condor	-0.0094	0.0037	0.0130	1.18	0.240	0.97	3541	4290
Risk Reversal	0.0241	0.0075	-0.0165	-0.73	0.464	1.01	2362	2860
Bull Call Spread	-0.0123	0.0008	0.0131	0.53	0.594	0.96	4724	5720
Call Ratio Spread	-0.0188	-0.0007	0.0181	0.78	0.435	1.03	4724	5720
Bear Put Spread	0.0190	-0.0066	-0.0255	-0.98	0.326	0.96	4067	5539
Put Ratio Spread	0.0381	-0.0004	-0.0386*	-1.77	0.077	0.96	4067	5539

Table 2: Structural-Break Check Around 2022 Daily-Expiration Expansion. The table reports pre/post means of strategy PNL relative to underlying (in percentage points), volatility ratios, and the post-period coefficient from regressions with combo fixed effects and date-clustered standard errors. The pre-period ends on April 14, 2022; the post-period starts on May 11, 2022; observations in between are excluded as transition.

Table 2 shows economically non-trivial shifts in several strategies' average PNL signs between the two regimes (for example, bull call and call ratio spreads improve, while bear put and put ratio spreads deteriorate), but these shifts are statistically weak once we account for strong within-day cross-combo dependence and heavy-tailed realizations. This evidence supports treating unconditional 0DTE strategy performance as regime-sensitive and motivates conditional rules rather than relying on full-sample means.

3.5 Implementability: Execution Costs, Turnover, and Capital Usage

The baseline results above are based on mid-quote marking. To evaluate implementability, we add execution friction at entry in three layers: (i) mid benchmark, (ii) bid/ask execution, where each strategy leg pays half of the observed option bid-ask spread, and (iii) bid/ask plus an additional 0.5 bp slippage-and-fee charge (in underlying-relative PNL units). We report strategy-level daily series at 10:00 ET, aggregating equally across moneyness combinations within each strategy.

Strategy	Mean Mid	Mean B/A	Mean B/A+0.5bp	SR Mid	SR B/A+0.5bp	Turnover (bps)	SR/Turnover	$ES_{1\%}$	Mean/$ES_{1\%}$	Obs
Strangle/Straddle	-0.0046	-0.0047	-0.0097	-0.20	-0.42	0.224	-1.86	1.2781	-0.008	1,319
Iron Butterfly/Condor	-0.0030	-0.0032	-0.0082	-0.24	-0.65	0.296	-2.21	0.5866	-0.014	1,319
Risk Reversal	0.0141	0.0140	0.0090	0.54	0.35	0.025	13.96	1.8215	0.005	1,319
Bull Call Spread	-0.0062	-0.0065	-0.0115	-0.22	-0.41	0.784	-0.53	0.8973	-0.013	1,319
Call Ratio Spread	-0.0105	-0.0108	-0.0158	-0.40	-0.60	0.745	-0.80	1.4357	-0.011	1,319
Bear Put Spread	0.0074	0.0072	0.0022	0.27	0.08	0.654	0.12	0.8256	0.003	1,318
Put Ratio Spread	0.0168	0.0165	0.0115	0.72	0.49	0.603	0.82	0.8228	0.014	1,318

Table 3: Implementable PNL Diagnostics at 10:00 ET. Means are in % of underlying. "B/A" subtracts half-spread execution cost computed from option-level bid-ask spreads and strategy legs. "B/A+0.5bp" further subtracts 0.5 basis points. Turnover is a gross entry premium proxy (in basis points of underlying), computed as absolute strategy premium plus half-spread cost. $ES_{1\%}$ is a 1% expected-shortfall proxy on daily strategy PNL under B/A+0.5bp.

Table 3 shows that moving from mid to implementable execution mechanically lowers mean PNLs and annualized Sharpe ratios for most strategies. The turnover proxy is small for all structures in underlying-relative terms, but capital usage measured by left-tail $ES_{1\%}$ remains large relative to average daily PNL, implying low mean-over-tail-capital ratios for most strategies. This reinforces that gross unconditional premiums can look materially better than implementable net performance, especially once friction and downside capital usage are accounted for.

3.6 Tail-Risk Diagnostics Beyond Mean PNL

Because strategy distributions are highly non-Gaussian, we explicitly report tail-focused risk diagnostics for implementable net PNL at 10:00 ET (half-spread plus 0.5bp slippage/fees).

Strategy	Skewness	$ES_{1\%}$	Max DD	Worst Day	Worst 5-Day	Loss Prob (%)	Obs
Strangle/Straddle	4.62	1.2781	23.475	-2.5113	-7.0941	70.4	1,319
Iron Butterfly/Condor	-0.47	0.5866	15.519	-0.7146	-1.6477	45.0	1,319
Risk Reversal	5.55	1.8215	8.209	-3.6411	-5.2603	36.5	1,319
Bull Call Spread	0.35	0.8973	21.473	-0.9755	-3.9473	54.8	1,319
Call Ratio Spread	-2.05	1.4357	26.453	-5.7459	-6.2393	53.5	1,319
Bear Put Spread	0.70	0.8256	13.285	-0.8641	-2.9679	61.5	1,318
Put Ratio Spread	0.47	0.8228	9.060	-2.1917	-2.6681	59.1	1,318

Table 4: Tail-Risk Diagnostics for Implementable 0DTE Strategy PNL. Daily PNL is in % of underlying and is net of half-spread and 0.5bp costs. Max drawdown is computed strategy-by-strategy as peak-to-trough decline of the cumulative daily net-PNL series (not a one-day loss and not a cross-strategy sum). $ES_{1\%}$, max drawdown, worst day, and worst 5-day outcomes emphasize downside exposure and distribution asymmetry.

Table 4 confirms that left-tail risk dominates mean effects for most strategies: $ES_{1\%}$ values are economically large (roughly 0.58–1.58% of underlying), worst-day outcomes are severe, and

cumulative drawdowns are substantial. Combined with mixed skewness signs across strategies, this reinforces that mean PNL alone is an insufficient summary for 0DTE strategy evaluation.

Taken together, these unconditional results are weak from an implementation perspective. Realized PNL is highly volatile, and the average variance premium is economically small at 0DTE horizons. Some favorable strategy outcomes may partly reflect the upward drift of the S&P 500 over our sample. We intentionally do not winsorize extremes, because rare tail realizations are central to the risk of these strategies. To illustrate time variation, Figure 5 plots 63-trading-

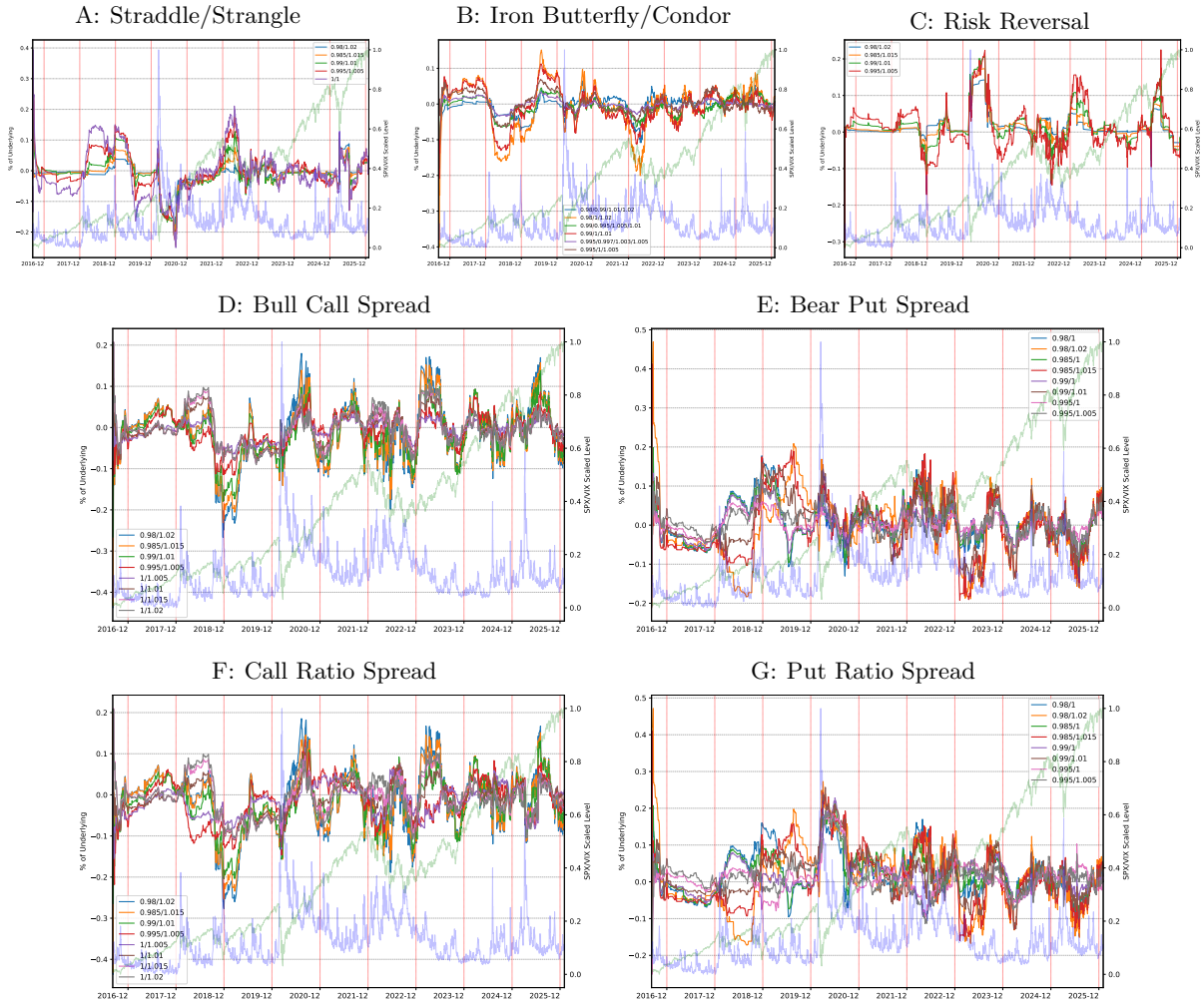


Figure 5: 0DTE Static Option Strategies: Time-series PNL. The figure shows 63-trading day moving average PNL of option strategies (from 10:00 ET to expiry at 16:00 ET) relative to underlying price $(payoff - mid\ price)/underlying\ price \times 100\%$. Secondary y-axis shows scaled to (0,1) series of SPX and VIX. The sample period is from 09/2016 to February 2, 2026.

day moving averages of strategy PNL together with scaled SPX and VIX series. No strategy

looks close to risk-free; profit and loss regimes can persist, but it is unclear whether they are identifiable *ex ante*. This motivates a more systematic look at what these strategies are actually loading on before we ask whether that variation can be forecast in real time.

4 What Drives Strategy PNL?

Before turning to implementable conditional rules, we examine which ex-post shocks drive daily strategy PNL. The candidate channels are not symmetric across structures. Variance-focused positions such as straddles and strangles should load more directly on realized variance, while directional spreads and skew trades should react more strongly to directional asymmetry and skewness realizations. We therefore relate strategy PNL at 10:00 ET to implied and realized moment measures, combining moneyness configurations within each strategy and absorbing mechanical differences across strike designs with combo fixed effects.

To quantify strategy sensitivities, we run strategy-specific pooled regressions of realized PNL on implied and realized moments (and related premium terms), combining multiple moneyness configurations within each strategy. Because changing strike configuration changes moment exposure, we include combo fixed effects.[5] Main results are reported in Table 5; last-hour analogs (15:00 ET to close) are in Appendix Table A4.

Panel A of Table 5 shows that realized variance alone explains little of cross-day strategy PNL. Even when RV coefficients are statistically significant, R^2 values are low (maximum about 4.5% for put ratio spreads). Directional spreads show opposite variance signs, consistent with negative index-variance comovement: higher variance tends to coincide with stronger down moves, helping bear structures and hurting bull structures. Adding realized skewness in Panel B materially improves fit for asymmetric strategies such as risk reversals and directional spreads.

[5] Allowing interactions between combo dummies and other regressors leaves the main conclusions intact. Adding Greeks directly as controls produced unstable and difficult-to-interpret estimates.

	Strangle	Irons	R/Reversal	C/Spread	C/R/Spread	P/Spread	P/R/Spread
Panel Realized Variance							
RV	-0.004	-0.001*	0.001	-0.004**	-0.002	0.004***	0.007***
	(-1.268)	(-1.918)	(0.295)	(-2.359)	(-0.894)	(2.901)	(5.676)
Combo FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2, Adj.	0.022	0.002	0.001	0.009	0.003	0.017	0.048
Obs.	5,342	6,409	4,274	8,548	8,548	7,728	7,728
Panel Realized Variance and Skewness							
RV	-0.005*	-0.001**	-0.006**	-0.010***	-0.004*	0.011***	0.010***
	(-1.794)	(-2.383)	(-2.083)	(-3.333)	(-1.733)	(3.636)	(3.954)
RS	0.019	0.006	0.137***	0.123***	0.044	-0.118***	-0.059***
	(0.787)	(1.323)	(6.087)	(5.276)	(1.569)	(-5.230)	(-3.258)
Combo FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2, Adj.	0.030	0.004	0.335	0.180	0.025	0.187	0.094
Obs.	5,342	6,409	4,274	8,548	8,548	7,728	7,728
Panel Implied and Realized Variances							
IV	-0.013**	0.004**	0.009	0.019***	0.021***	-0.021***	-0.010**
	(-2.274)	(2.124)	(1.256)	(3.093)	(4.297)	(-3.075)	(-2.542)
RV	0.013	-0.006**	-0.011	-0.028***	-0.028***	0.030***	0.020***
	(1.362)	(-2.171)	(-1.190)	(-3.773)	(-3.696)	(3.672)	(4.086)
Combo FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2, Adj.	0.056	0.009	0.014	0.050	0.049	0.064	0.060
Obs.	5,337	6,403	4,270	8,540	8,540	7,720	7,720
Panel Implied and Realized Variances and Skewness							
IV	-0.016**	0.005**	0.003	0.019***	0.024***	-0.021***	-0.012***
	(-2.326)	(2.038)	(0.446)	(3.274)	(4.478)	(-3.475)	(-3.308)
IS	-0.008	0.008	0.028*	0.064***	0.053***	-0.069***	-0.052***
	(-0.375)	(1.084)	(1.788)	(5.291)	(2.842)	(-5.066)	(-4.531)
RV	0.013	-0.006**	-0.007	-0.025***	-0.028***	0.028***	0.019***
	(1.349)	(-2.241)	(-0.848)	(-4.436)	(-4.123)	(4.755)	(4.239)
RS	0.021	0.011	0.157***	0.155***	0.065***	-0.157***	-0.090***
	(0.830)	(1.577)	(7.384)	(8.832)	(2.936)	(-8.912)	(-6.396)
Combo FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2, Adj.	0.071	0.013	0.350	0.233	0.089	0.263	0.131
Obs.	5,337	6,403	4,270	8,540	8,540	7,720	7,720

Table 5: Option Strategies PNL vs. Implied and Realized Moments. The table shows the results of regressing realized PNL of option strategies (10:00 ET to expiry at 16:00 ET) on implied and realized distribution moments. PNL is specified per one unit of underlying relative to underlying price $(payoff - mid\ price)/underlying\ price \times 100\%$. The result in each column is based on a pooled regression of strategy PNL for several moneyness combinations, including combo fixed effects (Combo FE) and date-clustered standard errors. The sample period is from 09/2016 to February 2, 2026.

Ratio-spread fit also improves, though sensitivity becomes unstable when large directional moves dominate.

Including implied variance and implied skewness proxies in Panels C and D adds further explanatory power, typically by 2–7 percentage points of R^2. The increase is strongest for strangles, directional spreads, and call ratio spreads. Overall, strategy PNL is driven more by realized skewness (directional movement) than by realized variance alone, except for variance-focused structures such as strangles. Relative to evidence for delta-hedged options (e.g., Büchner and Kelly 2022), the link to variance premia is weaker and less predictable in our unhedged 0DTE setting. Last-hour regressions in Appendix Table A4 amplify this pattern: fits are often higher, and realized skewness remains the dominant channel.

4.1 Inference Robustness: Date Clustering and Multiple Testing

As an inference-robustness step, we re-estimate the key pooled regressions at 10:00 ET with combo fixed effects and date-clustered standard errors, and then adjust p-values for multiple testing using Benjamini-Hochberg FDR control (Benjamini and Hochberg 1995), following concerns about repeated testing in predictability settings (Harvey, Liu, and Zhu 2015).

Table 6 shows that the strongest directional-skew links remain after clustering and BH adjustment, while many weaker coefficients lose support. This sharpens the main interpretation of the unconditional evidence: what matters for many 0DTE strategies is not generic exposure to same-day variance, but directional asymmetry and the skewness channel. The conditional question is therefore whether states observed by 10:00 ET can capture enough of that variation to improve trading decisions in real time.

Strategy	Spec	Coef (key var)	t (clustered)	p-value	q-value (BH)	Obs
Strangle/Straddle	RV only	-0.004	-1.30	0.194	0.239	5,337
Strangle/Straddle	RV + RS	0.018	0.71	0.476	0.500	5,337
Strangle/Straddle	IV + IS + RV + RS	0.021	0.83	0.407	0.449	5,337
Iron Butterfly/Condor	RV only	-0.001	-1.81	0.070	0.113	6,403
Iron Butterfly/Condor	RV + RS	0.008	1.71	0.087	0.130	6,403
Iron Butterfly/Condor	IV + IS + RV + RS	0.011	1.58	0.115	0.161	6,403
Risk Reversal	RV only	0.001	0.28	0.777	0.777	4,270
Risk Reversal	RV + RS	0.139	6.09	0.000	0.000	4,270
Risk Reversal	IV + IS + RV + RS	0.157	7.38	0.000	0.000	4,270
Bull Call Spread	RV only	-0.004	-2.55	0.011	0.019	8,540
Bull Call Spread	RV + RS	0.121	5.21	0.000	0.000	8,540
Bull Call Spread	IV + IS + RV + RS	0.155	8.83	0.000	0.000	8,540
Call Ratio Spread	RV only	-0.002	-0.98	0.326	0.380	8,540
Call Ratio Spread	RV + RS	0.041	1.48	0.140	0.184	8,540
Call Ratio Spread	IV + IS + RV + RS	0.065	2.94	0.003	0.007	8,540
Bear Put Spread	RV only	0.004	2.89	0.004	0.007	7,720
Bear Put Spread	RV + RS	-0.120	-5.19	0.000	0.000	7,720
Bear Put Spread	IV + IS + RV + RS	-0.157	-8.91	0.000	0.000	7,720
Put Ratio Spread	RV only	0.007	5.66	0.000	0.000	7,720
Put Ratio Spread	RV + RS	-0.060	-3.24	0.001	0.003	7,720
Put Ratio Spread	IV + IS + RV + RS	-0.090	-6.40	0.000	0.000	7,720

Table 6: Clustered-SE Inference with BH Multiple-Testing Adjustment. Reported coefficients are from pooled regressions with combo fixed effects and date-clustered standard errors; q-values are Benjamini-Hochberg adjustments across all reported tests.

5 Conditional Trading Rules

The conditional exercise asks whether these state links can be turned into implementable real-time rules. The answer is not uniform across strategies, but neither is it uniformly negative. Once the problem is cast as a directional classification task and estimated under strict no-look-ahead windows, several strategy-specific rules and diversified baskets deliver meaningful net out-of-sample performance.

All strategies in this paper are unhedged and held to expiration. Their PNL is therefore driven by entry cost and by the realized underlying move from entry to close. Entry costs are observable and related to expected variance, but forecasting the *signed* daily index move is notoriously difficult.[6] A more realistic target is directionally asymmetric intraday payoffs and higher-moment variation rather than fine-grained return magnitudes. This is why the

[6]Our baseline view is that short-horizon directional predictability in the index is economically weak.

conditional analysis below treats direction prediction, probability calibration, and net economic value as separate objects.

5.1 A Simple State Filter at 10:00 ET: VIX Regimes (Ex-Post Cutoffs)

As a minimal implementable conditioning step, we sort trading days into low/mid/high regimes using full-sample terciles of the 10:00 ET 0DTE implied-variance measure (from our VIX-style estimator). Concretely, this variable is the integrated implied variance from 10:00 ET to end-of-day (16:00 ET), constructed from SPXW 0DTE options using the VIX methodology. Regime thresholds are therefore identified *ex post* (full-sample terciles), while the state variable itself is observed *ex ante* at entry on each day. We then compute strategy PNL means across these regimes.

Strategy	Mean Full	Mean Low-VIX	Mean Mid-VIX	Mean High-VIX	High-Low	t(H-L)	p(H-L)	Obs
Strangle/Straddle	-0.0046	-0.0006	-0.0032	-0.0100	-0.0095	-0.33	0.745	1,319
Iron Butterfly/Condor	-0.0030	0.0016	0.0005	-0.0111	-0.0127	-0.93	0.353	1,319
Risk Reversal	0.0141	-0.0016	-0.0043	0.0482	0.0498	1.49	0.136	1,319
Bull Call Spread	-0.0062	-0.0035	-0.0171	0.0019	0.0054	0.17	0.863	1,319
Call Ratio Spread	-0.0105	-0.0027	-0.0134	-0.0155	-0.0127	-0.42	0.673	1,319
Bear Put Spread	0.0074	-0.0024	0.0134	0.0113	0.0137	0.44	0.660	1,318
Put Ratio Spread	0.0168	-0.0036	0.0119	0.0420	0.0455	1.75	0.080	1,318

Table 7: Conditional Strategy PNL by VIX Regime at 10:00 ET (Ex-Post Tercile Cutoffs). Means are daily strategy PNL in % of underlying (10:00 ET to 16:00 ET), with equal weighting across moneyness combinations within each strategy and day. Regimes are based on full-sample terciles of the 10:00 ET integrated implied variance (10:00–16:00) from SPXW 0DTE options, constructed with the VIX methodology.

Table 7 shows clear regime heterogeneity: downside-oriented structures (bear put and put ratio spreads) earn substantially higher average PNL in high-VIX states than in low-VIX states, while upside-oriented call-spread structures perform better in low-VIX states. Economically, this suggests that even simple state partitions can alter unconditional performance profiles materially, though statistical strength remains limited for most strategies in this static design.

5.2 Strict Out-of-Sample Conditional Protocol

To keep the conditioning exercise implementable, we use a strict out-of-sample protocol with no look-ahead construction of features. On each date, predictors include only variables observed by 10:00 ET: contemporaneous implied/state quantities (including the 10:00 ET SPXW 0DTE integrated implied variance from 10:00 to 16:00, plus skew/slope proxies), lagged realized variables, and lagged strategy-PNL terms. We estimate both expanding-window and rolling-window schemes (252 trading-day minimum) for sign prediction of net strategy PNL and evaluate them using hit rate, probability calibration, and net economic value after execution costs. With this burn-in, OOS forecasts begin in April 2019 and run through February 2, 2026. This design follows the practical OOS discipline emphasized by Goyal and Welch (2008) and aligns with the event-state perspective in Knox, Londono, Samadi, and Vissing-Jorgensen 2025.

To avoid pooling many overlapping strike designs in one conditional exercise, we fix one representative near-ATM moneyness configuration for each strategy type. Candidate configurations satisfy $\max_l |M_l - 1| \leq 0.01$; within each strategy we select the most frequent configuration by trading-day coverage. The selected configurations are listed in Table 8 and all lie within $\pm 0.5\%$ of ATM.

Strategy	Representative moneyness	Days	Max $\lvert M-1\rvert$
Bear Put Spread	0.995/1	1318	0.005
Bull Call Spread	1/1.005	1319	0.005
Call Ratio Spread	1/1.005	1319	0.005
Iron Butterfly/Condor	0.995/0.997/1.003/1.005	1319	0.005
Put Ratio Spread	0.995/1	1318	0.005
Risk Reversal	0.995/1.005	1319	0.005
Strangle/Straddle	0.995/1.005	1319	0.005

Table 8: Representative Strategy Moneyness Configurations for Conditional Tests. For each strategy type, we select one near-ATM configuration under $\max_l |M_l - 1| \leq 0.01$, choosing the configuration with the highest day coverage in the sample. The last column reports the maximum across legs of the absolute distance from ATM, $\max_l |M_l - 1|$, not the width of the full structure. Thus, a configuration such as 0.995/0.997/1.003/1.005 has Max $|M - 1| = 0.005$ because its most extreme legs are 0.5% away from ATM.

After this representative-moneyness restriction, we keep the main-text conditional evidence fully strategy-specific to avoid additional averaging across strategy panels. The benchmark specifications in Tables 10 and 11 use a deliberately parsimonious predictor set: market-state variables plus lagged strategy-PNL terms. Richer baseline/GEX/flow/liquidity stacks are examined separately in the model-zoo exercises, where GEX variables are flow/exposure proxies based on traded volume, open interest, and leg Greeks rather than a dealer-inventory reconstruction. Define net strategy return as

$$y_{s,t}^{net} = PNL_{s,t}^{net} = reth_und_{s,t} - c_{s,t}, \tag{29}$$

where $c_{s,t}$ is implementation cost (half-spread plus fixed 0.5bp).

We compare three prediction-to-position designs under one common OOS setup:

$$\textbf{(A) Return target:} \quad \hat{y}_{s,t} = f_s(X_{s,t}), \qquad w_{s,t}^{(R)} = \text{sign}(\hat{y}_{s,t}), \tag{30}$$

$$\textbf{(B) Binary target (Logistic):} \quad p_{s,t} = \Pr(y_{s,t}^{net} > 0 \mid X_{s,t}) = \Lambda(\alpha_s + \beta_s^\top X_{s,t}), \tag{31}$$

$$\log\left(\frac{p_{s,t}}{1-p_{s,t}}\right) = \alpha_s + \beta_s^\top X_{s,t}, \tag{32}$$

$$\textbf{(B1) Hard map:} \quad w_{s,t}^{(H)} = \text{sign}(p_{s,t} - 0.5) \in \{-1, +1\}, \tag{33}$$

$$\textbf{(B2) Soft map:} \quad w_{s,t}^{(S)} = 2p_{s,t} - 1 \in [-1, +1]. \tag{34}$$

For all three designs, realized trading return is $r_{s,t} = w_{s,t}\, y_{s,t}^{net}$. In words, hard mapping is full-size directional timing, while soft mapping is confidence-weighted directional timing.

The economic difference is simple: (A) forecasts payoff magnitude and then trades direction; (B1) forecasts only direction and trades full size; (B2) forecasts direction and scales position size by confidence. In our implementation, each strategy is estimated separately at 10:00 ET using only information available by that time (contemporaneous implied-state predictors and lagged realized predictors), with no-look-ahead feature construction.

We keep logistic as the benchmark in strategy-level tests for three reasons. First, the trading decision is directional (long vs short), so modeling $\Pr(y^{net} > 0)$ is directly aligned with the traded object. Second, logistic outputs are bounded probabilities, which allows transparent calibration and Brier-score diagnostics in addition to hit rates. Third, in short-horizon 0DTE data with noisy payoffs and limited per-strategy sample sizes, a low-variance parametric benchmark is less prone to unstable overfitting than higher-capacity nonlinear models.

Model Family	SR (Return)	Mean (Return, bps)	SR (Hard)	Mean (Hard, bps)	SR (Soft)	Mean (Soft, bps)
Ridge	-0.34	-0.319	1.07	0.891	0.84	0.352
Elastic Net	0.34	0.305	1.06	0.894	0.92	0.385
Random Forest	-0.43	-0.409	0.60	0.547	0.19	0.059
LightGBM	0.48	0.485	0.13	0.127	0.35	0.155
XGBoost	-0.32	-0.313	0.65	0.606	0.47	0.183

Table 9: Target Choice Experiment: Return Prediction vs Logistic Hard/Soft Mapping. Each row compares the same model family under three implementations on the same representative stacked OOS panel: we first estimate strategy-specific models separately (one representative moneyness per strategy), then stack resulting strategy-day OOS returns across strategies and dates. Thus, pooling here is at the evaluation stage (stacked strategy-day observations), not a single cross-strategy model fit. Implementations are: (A) direct return prediction with $w_t = \text{sign}(\hat{y}_t)$, (B1) logistic binary prediction with hard mapping $w_t = \text{sign}(\hat{p}_t - 0.5)$, and (B2) logistic binary prediction with soft mapping $w_t = 2\hat{p}_t - 1$. Columns report annualized SR net and mean net PNL (bps).

Construction details for Table 10. Let s index strategy and t index date. For each strategy s, we estimate a separate logistic model on that strategy's own daily series only (no cross-strategy pooling): $y_{s,t} = \mathbf{1}\{PNL^{net}_{s,t} > 0\}$ with predictors observed by 10:00 ET. In the benchmark, these predictors are the 10:00 ET implied-state variables $(IV_t, IS_t, slope_up_t, slope_dn_t)$, one-day-lagged realized SPX moments, and lagged strategy-PNL terms $(P^{(1)}, \bar{P}^{(5)}, \sigma^{(5)}_P)$. After representative-moneyness selection (Table 8), each strategy contributes one moneyness configuration; we do not average across multiple strike designs in this table. At each OOS date t, estimation uses data through $t-1$ only, with either expanding or 252-day rolling windows; predictors are standardized within the training window, and the logistic model is estimated with L2 regularization. We then form $\hat{p}_{s,t}$, trade with $\text{sign}(\hat{p}_{s,t} - 0.5)$, and report OOS diagnostics.

Table 9 shows a clear ranking of target designs. Direct return prediction remains weak in this short-horizon setting: only the Elastic Net and LightGBM regressions stay modestly positive

Strategy	Protocol	Hit Rate (%)	Brier	Calib. Slope	Mean Net (bps)	SR Net	N
Strangle/Straddle	Expanding (252d)	71.8	0.197	0.38	2.171	0.70	1,061
Strangle/Straddle	Rolling (252d)	70.2	0.200	0.24	2.157	0.70	1,061
Iron Butterfly/Condor	Expanding (252d)	63.9	0.227	0.68	0.446	0.82	1,061
Iron Butterfly/Condor	Rolling (252d)	63.1	0.234	0.44	0.442	0.82	1,061
Risk Reversal	Expanding (252d)	74.5	0.183	0.47	-0.374	-0.09	1,061
Risk Reversal	Rolling (252d)	74.8	0.183	0.49	1.531	0.37	1,061
Bull Call Spread	Expanding (252d)	57.9	0.245	-0.01	0.276	0.21	1,061
Bull Call Spread	Rolling (252d)	56.5	0.252	-0.04	0.260	0.20	1,061
Call Ratio Spread	Expanding (252d)	58.9	0.245	0.31	-0.319	-0.13	1,061
Call Ratio Spread	Rolling (252d)	59.4	0.247	0.16	-0.538	-0.22	1,061
Bear Put Spread	Expanding (252d)	64.0	0.233	0.01	0.473	0.36	1,060
Bear Put Spread	Rolling (252d)	62.2	0.239	-0.05	0.431	0.33	1,060
Put Ratio Spread	Expanding (252d)	67.1	0.225	0.93	2.578	1.26	1,060
Put Ratio Spread	Rolling (252d)	65.7	0.225	0.57	2.159	1.05	1,060

Table 10: Strict Out-of-Sample Conditional Performance at 10:00 ET (Representative Moneyness). Each strategy is estimated separately with a logistic benchmark on the representative moneyness configuration (Table 8), using only 10:00 ET information: contemporaneous implied/state variables, lagged realized variables, and lagged strategy-PNL terms. No cross-strategy pooling is used in this table; each row is one strategy under one protocol. We report two window rules (expanding and rolling, minimum 252 trading days). Directional signal is $\text{sign}(\hat{p}_t - 0.5)$ and net daily strategy return is this sign times strategy PNL net of half-spread and 0.5bp. OOS forecasts begin after the 252-day burn-in in April 2019 and run through February 2, 2026. Reported columns are hit rate (%), Brier score, calibration slope, mean net PNL (bps), annualized SR net, and OOS observations.

after costs, while the other return-target specifications are negative. By contrast, binary designs are materially stronger. Within the binary setup, hard mapping dominates soft mapping in four of the five model families once costs are applied, with the ridge-logit and elastic-net-logit variants reaching SRs slightly above 1.0. We therefore keep binary-hard as the main implementation and treat soft mapping as robustness. For strategy-level tests, we retain the logistic benchmark because it remains stable across mappings and provides transparent probability diagnostics.

Table 10 shows that the fixed logistic benchmark delivers economically meaningful strategy-specific OOS results under both the expanding and rolling protocols. Put ratio spreads are the strongest individual series, with 2.58 bps mean net and an SR of 1.26 in the expanding specification. Strangle/straddle structures remain positive, although more modest once the 2025–2026 extension is included. Iron butterfly/condor structures still retain an attractive Sharpe ratio of 0.82. Other strategies are weaker, and call ratio spreads remain unattractive. Conditioning therefore helps, but selectively rather than uniformly.

5.3 OOS Investment Time-Series from Model Signals

To translate Table 10 into an investable object, we construct daily OOS strategy returns from the same strategy-specific logistic benchmark signals using a common rolling-window estimation rule (252-trading-day window) for all strategies. The strategy signal is then applied to the same representative moneyness configuration and net-PNL definition as in Table 10.

Strategy	Mean Net (bps)	SR Net	Hit Rate (%)	Long Share (%)	$ES_{1\%}$ (bps)	Worst Day (bps)	Max DD (bps)	Days
Put Ratio Spread	2.159	1.05	65.7	31.2	-153.12	-354.21	-531.85	1060
Strangle/Straddle	2.157	0.70	70.2	8.3	-266.60	-601.07	-1341.07	1061
Risk Reversal	1.531	0.37	74.8	87.8	-283.04	-535.16	-2006.84	1061
Iron Butterfly/Condor	0.442	0.82	63.1	47.8	-16.48	-17.46	-204.21	1061
Bear Put Spread	0.431	0.33	62.2	8.9	-38.86	-43.53	-479.78	1060
Bull Call Spread	0.260	0.20	56.5	15.8	-35.23	-36.64	-619.43	1061
Call Ratio Spread	-0.538	-0.22	59.4	26.7	-250.45	-690.00	-1173.55	1061
Equal-weight basket (Top 3 by Mean PNL)	1.949	1.17	69.4	42.5	-121.66	-206.83	-520.91	1061
Equal-weight basket (Top 3 by SR)	1.587	1.27	61.3	29.1	-99.91	-224.84	-474.91	1061
Equal-weight basket (All)	0.920	1.01	59.8	32.4	-65.24	-139.08	-392.96	1061

Table 11: OOS Investment Performance from Strategy-Specific Signals (Rolling Window). For each strategy, we use the same logistic benchmark as in Table 10, estimated with a 252-trading-day rolling window. Daily strategy return is $\text{sign}(\hat{p}_t - 0.5) \times$net strategy PNL for the representative moneyness configuration. OOS forecasts begin after the 252-day burn-in in April 2019 and run through February 2, 2026. The table reports all strategy series and equal-weight baskets (top-three by mean net PNL, top-three by SR net, and all-strategies), with columns for mean net (bps), annualized SR net, hit rate (%), long share (%), $ES_{1\%}$, worst day, max drawdown, and days.

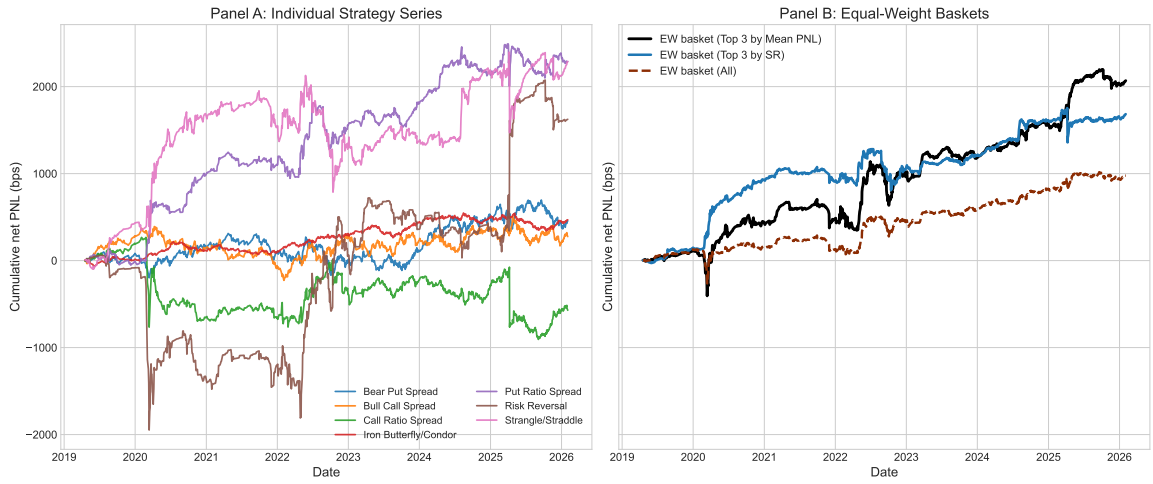


Figure 6: Cumulative OOS Net PNL from Strategy-Specific Signals. Panel A plots cumulative net PNL (bps) for all individual strategy series. Panel B plots three equal-weight baskets built from the same strategy-level series: top-three selected by mean net PNL, top-three selected by SR net, and all-strategies basket. Each strategy uses the same logistic benchmark as in Table 10, estimated with a 252-trading-day rolling window.

Table 11 and Figure 6 show the strategy-specific implementation directly implied by Table 10. Diversification still matters in the extended sample. The top-three basket ranked by SR net reaches a net Sharpe ratio of 1.27, the top-three basket ranked by mean net PNL reaches 1.17,

and even the all-strategies basket reaches 1.01. Equal weighting does not eliminate tail risk, but it materially smooths strategy-specific drawdowns and makes the conditional evidence easier to interpret as a tradable object rather than a collection of isolated strategy series.

6 Conclusion

This paper studies realized payoff distributions of 0DTE SPXW options and common multi-leg strategy templates over 09/2016 to February 2, 2026. A positive 0DTE variance risk premium exists, but at same-day horizons its economic magnitude is modest and it is not the central object from an implementation perspective. Once attention shifts from a synthetic variance object to actual option positions and feasible multi-leg structures, the main empirical fact is the wide and state-dependent distribution of realized outcomes rather than a single unconditional average edge.

The dominant message of the unconditional evidence is risk. Strategy-level distributions are dispersed, asymmetric, and fragile across regimes. After execution costs, downside risk remains large relative to mean carry: Table 4 reports substantial $ES_{1\%}$ values, severe worst-day outcomes, and meaningful cumulative drawdowns even for strategies with positive average PNL. Mean PNL on its own is therefore an inadequate summary statistic for 0DTE implementation.

At the same time, the conditional evidence is stronger than a purely pessimistic reading would suggest. The main economic link runs through directional asymmetry and skewness rather than through realized variance alone (Tables 5 and 6). When the forecasting problem is posed as directional classification and evaluated under strict out-of-sample discipline, several strategy-specific rules remain attractive after costs. In the updated benchmark, put ratio spreads reach a net Sharpe ratio of 1.26 and iron butterfly/condor structures 0.82 in Table 10; at the portfolio level, the top-three-by-SR basket reaches 1.27 and the all-strategies basket reaches 1.01

in Table 11. The evidence is therefore selective rather than diffuse: some strategy families can be timed meaningfully, many others cannot.

The main limitation of the paper is scope. We study SPXW 0DTE positions held to the close, without dynamic intraday hedging and without a causal market-impact design. Our conditioning variables are restricted to information available by 10:00 ET, which is intentional but demanding. For that reason, the natural next step is not to search for one universal 0DTE rule, but to continue updating the data under locked out-of-sample governance, extend event conditioning, and monitor whether the better-performing strategy families remain stable as the market evolves.

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A Additional Tables and Figures

A.1 Strategy Templates

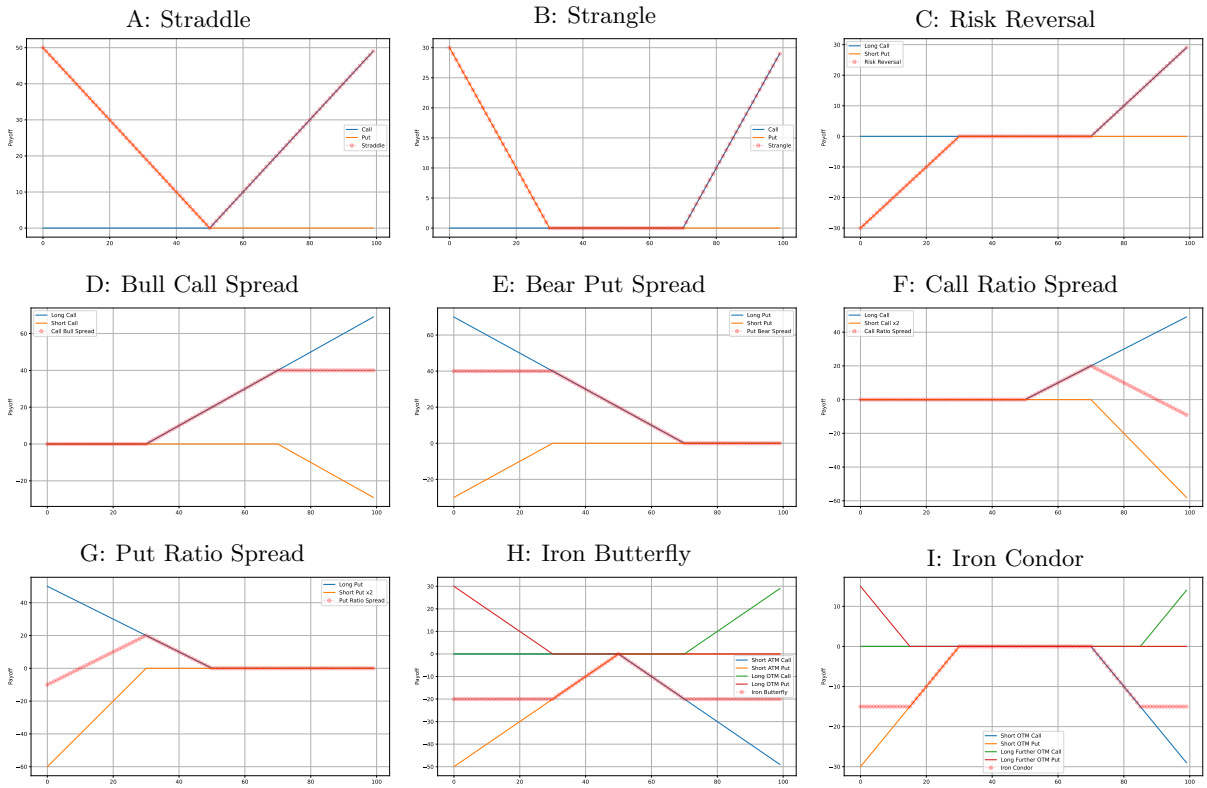


Figure A1: Sample Payoffs of Option Strategies. The figure shows illustrative terminal payoffs for the strategy templates used in the paper. Payoffs exclude entry premium. Converting payoff to PNL shifts debit structures downward by the paid premium and credit structures upward by the received premium.

A.2 Unconditional Trading Rules

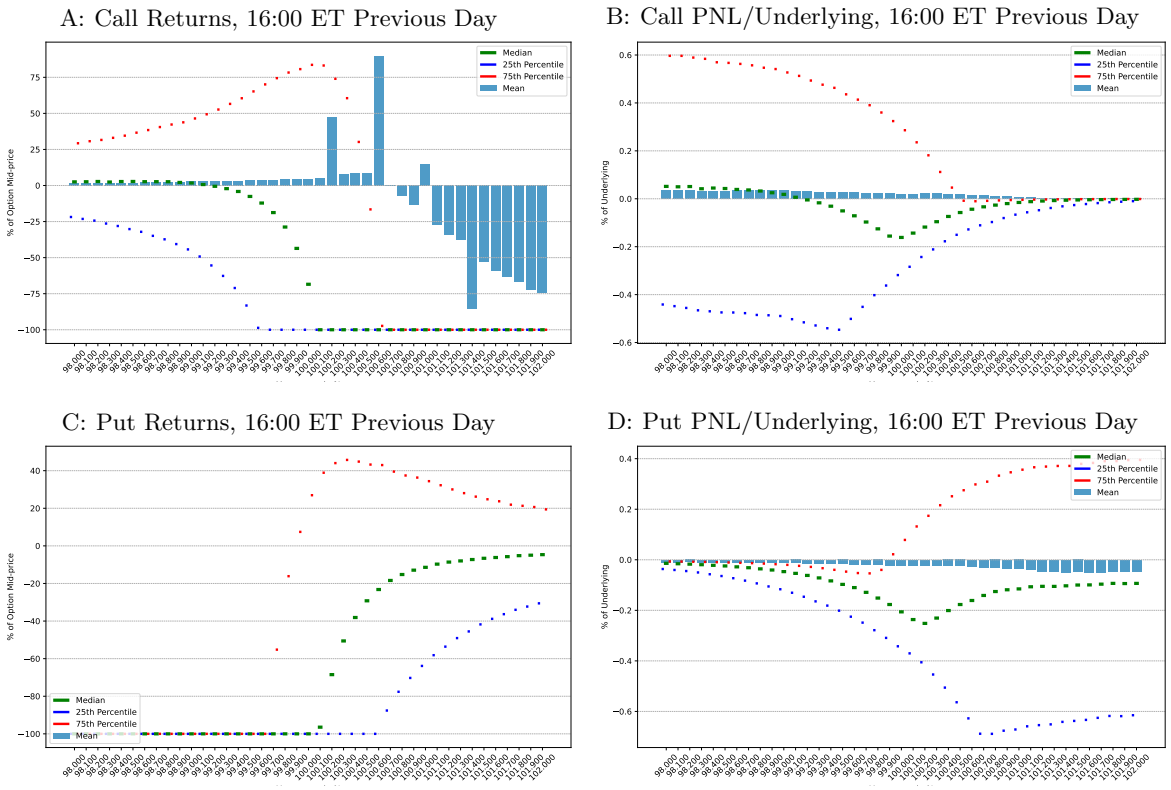


Figure A2: Option Returns, 16:00 ET on the Previous Day to Next-Day Expiry. The figure provides statistics on profitability of call and put option buying at 16:00 ET on the previous day and holding to expiry at 16:00 ET on the next day. These contracts are one day from expiry at entry and become 0DTE on the following morning. Panels on the left show realized return in % relative to option mid-price. Panels on the right show option realized profit (payoff - mid-price) per one unit of underlying relative to underlying price, in %. Bars show mean values, and each bar is accompanied by median, 25th and 75th percentiles. X-axis labels show moneyness of the analyzed options. The sample period is from 09/2016 to February 2, 2026.

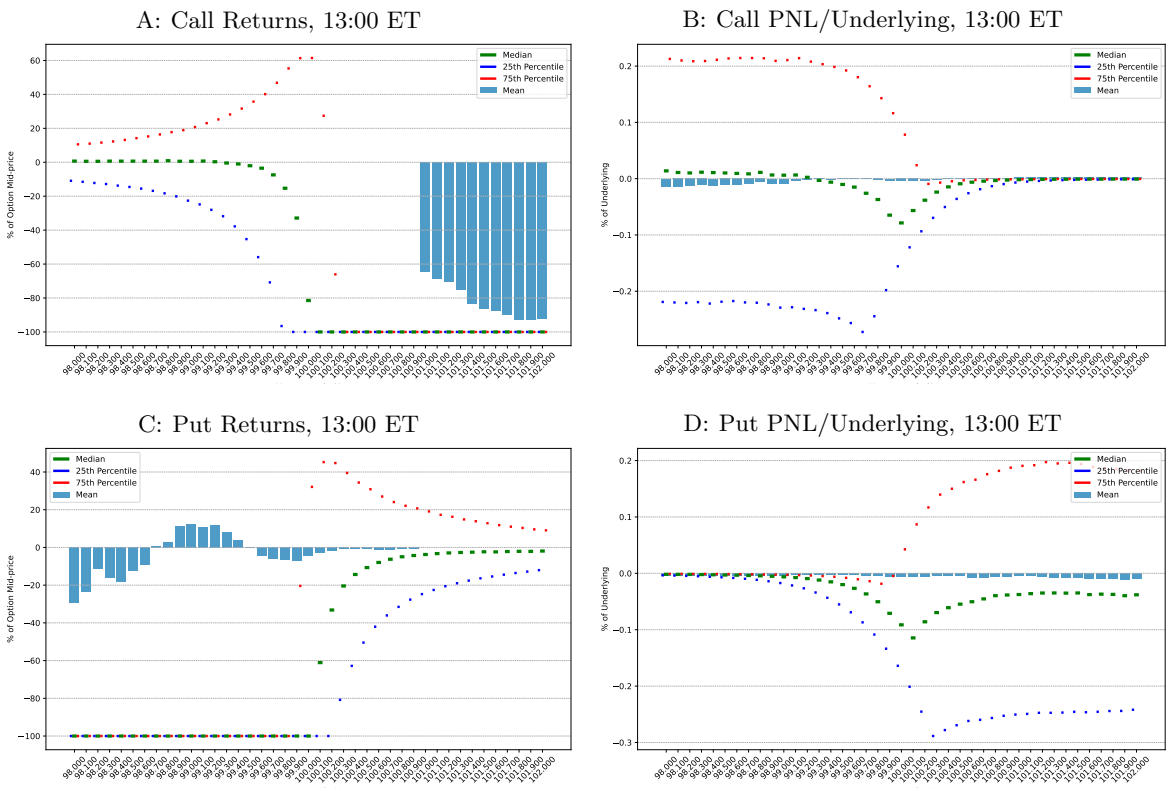


Figure A3: 0DTE Option Returns, 13:00 ET to Expiry. The figure provides statistics on profitability of naked 0DTE call and put option buying at 13:00 ET and holding to expiry at 16:00 ET. Panels on the left show realized return in % relative to option mid-price. Panels on the right show option realized profit (payoff - mid-price) per one unit of underlying relative to underlying price, in %. Bars show mean values, and each bar is accompanied by median, 25th and 75th percentiles. X-axis labels show moneyness of the analyzed options. The sample period is from 09/2016 to February 2, 2026.

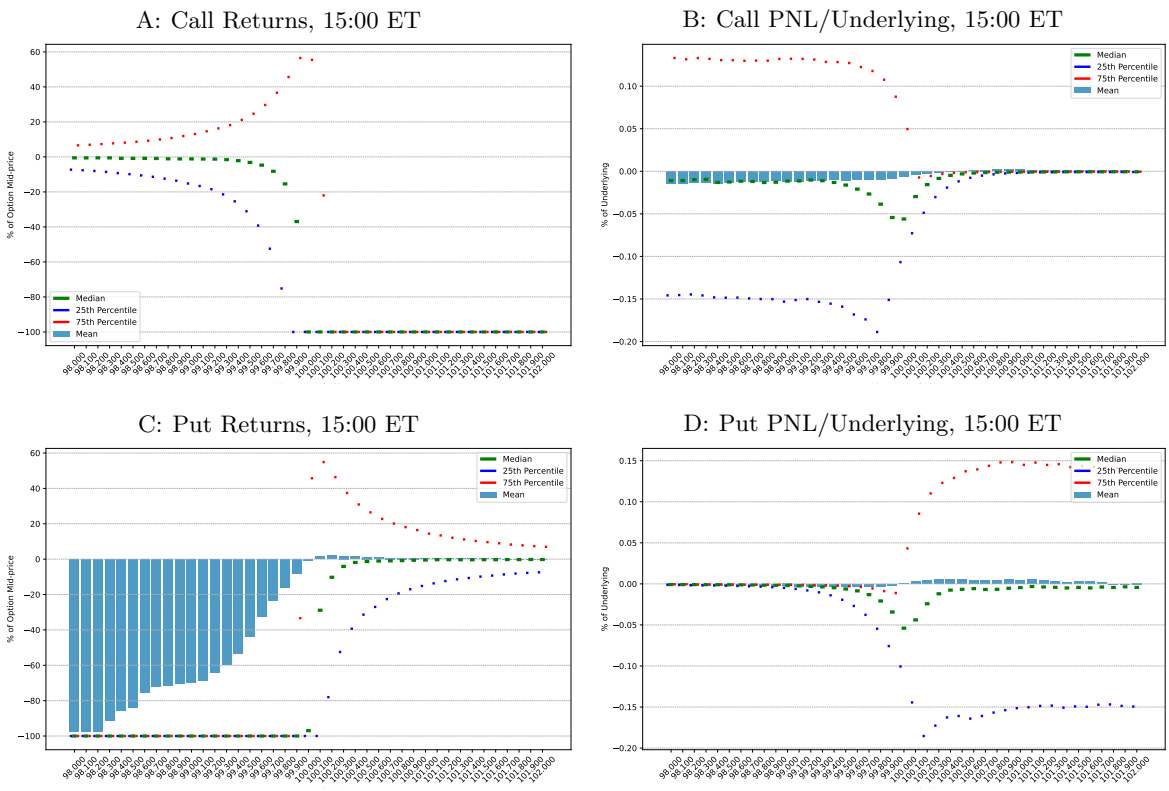


Figure A4: 0DTE Option Returns, 15:00 ET to Expiry. The figure provides statistics on profitability of naked 0DTE call and put option buying at 15:00 ET and holding to expiry at 16:00 ET. Panels on the left show realized return in % relative to option mid-price. Panels on the right show option realized profit (payoff - mid-price) per one unit of underlying relative to underlying price, in %. Bars show mean values, and each bar is accompanied by median, 25th and 75th percentiles. X-axis labels show moneyness of the analyzed options. The sample period is from 09/2016 to February 2, 2026.

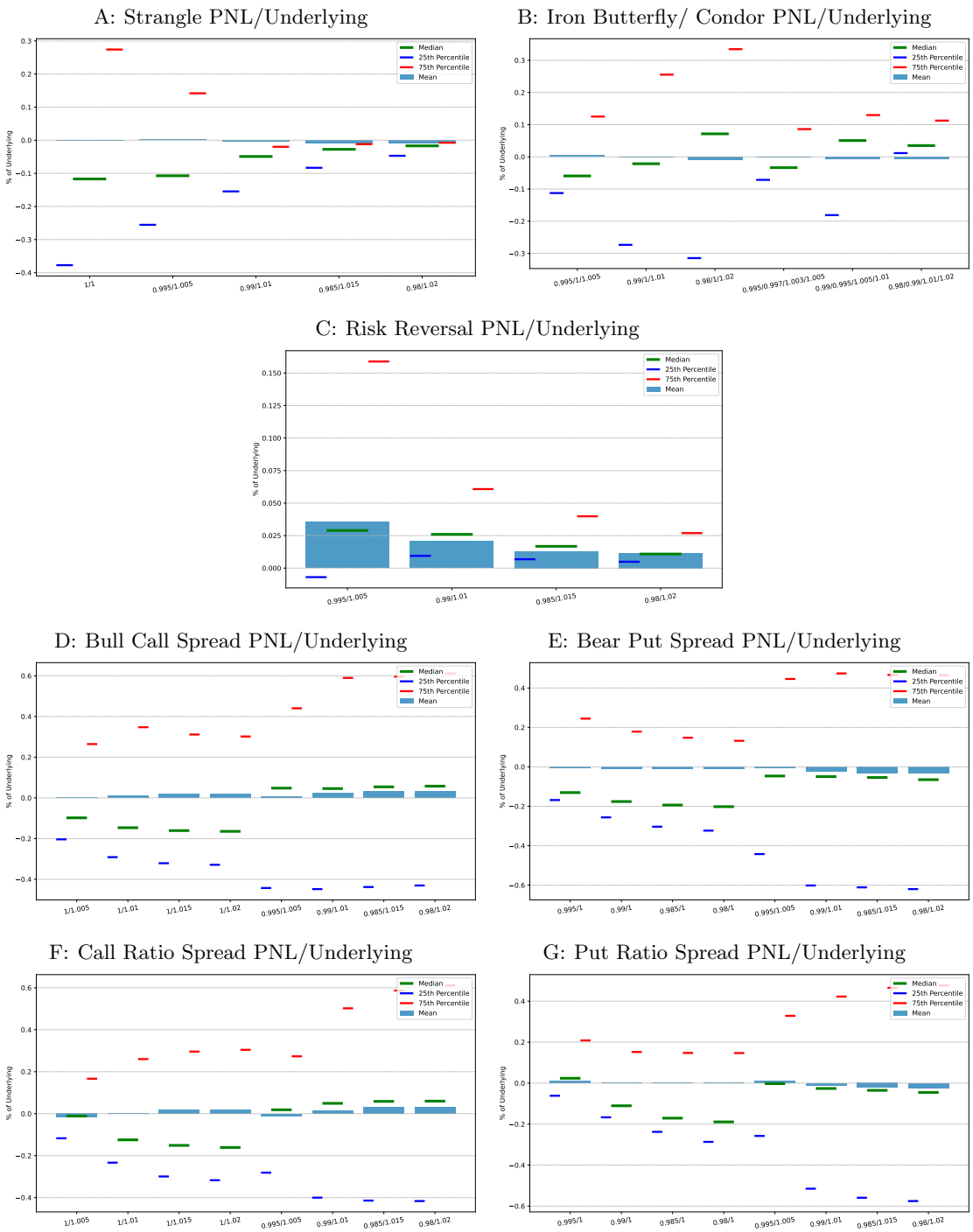


Figure A5: Static Option Strategies, 16:00 ET on the Previous Day to Next-Day Expiry. The figure provides statistics on profitability of several well-known option strategies based on short-dated call and put options; positions are taken at 16:00 ET on the previous day and held to expiry at 16:00 ET on the next day. These positions are one day from expiry at entry. All panels show option strategy realized profit (payoff - mid-price) per one unit of underlying relative to the underlying price, in %. Bars show mean values, and each bar is accompanied by median, 25th and 75th percentiles. X-axis labels show combination of moneyness for options used for a strategy. The sample period is from 09/2016 to February 2, 2026.

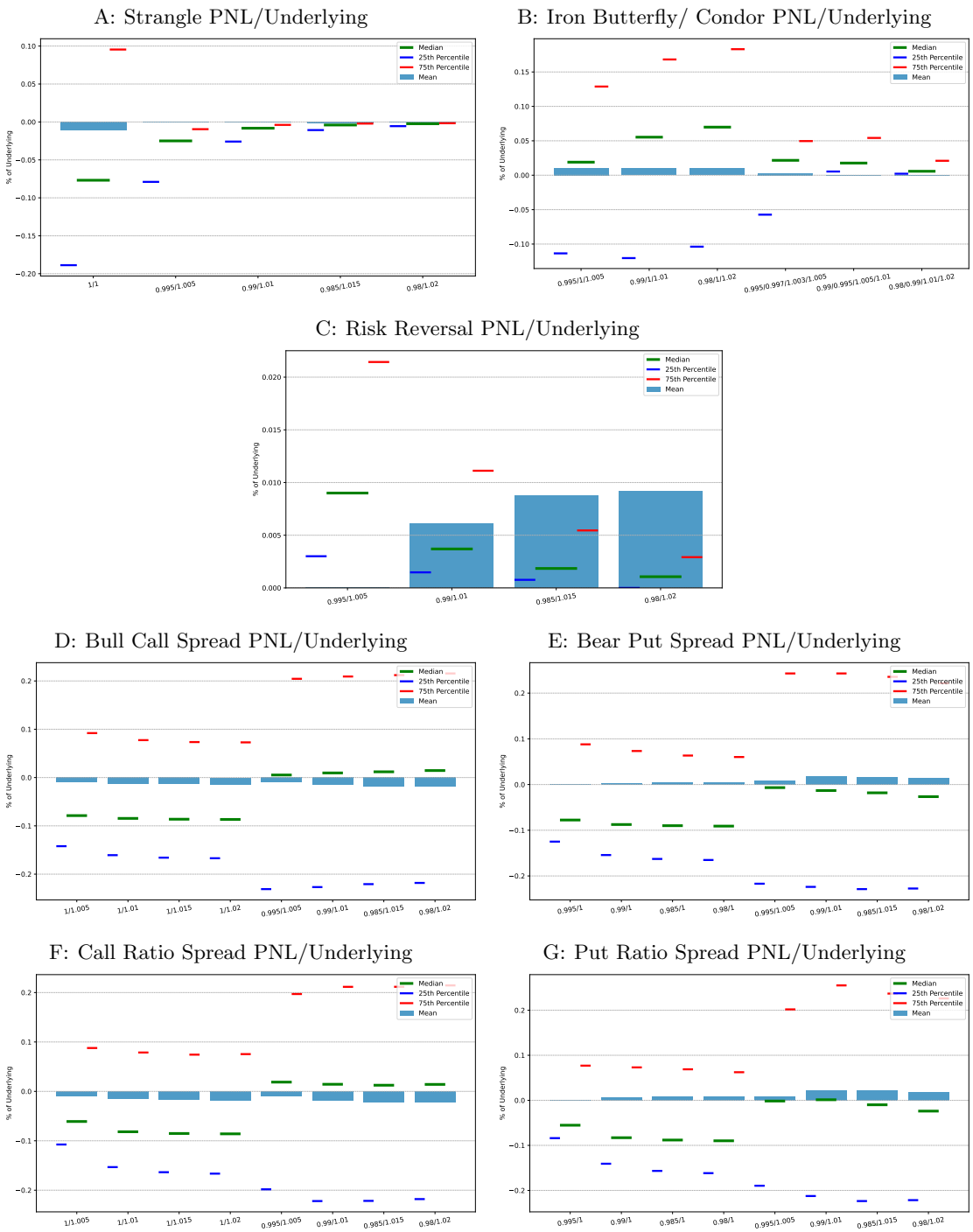


Figure A6: 0DTE Static Option Strategies, 13:00 ET to Expiry. The figure provides statistics on profitability of several well-known option strategies based on 0DTE call and put options; positions are taken at 13:00 ET and held to expiry at 16:00 ET. All panels show option strategy realized profit (payoff - mid-price) per one unit of underlying relative to the underlying price, in %. Bars show mean values, and each bar is accompanied by median, 25th and 75th percentiles. X-axis labels show combination of moneyness for options used for a strategy. The sample period is from 09/2016 to February 2, 2026.

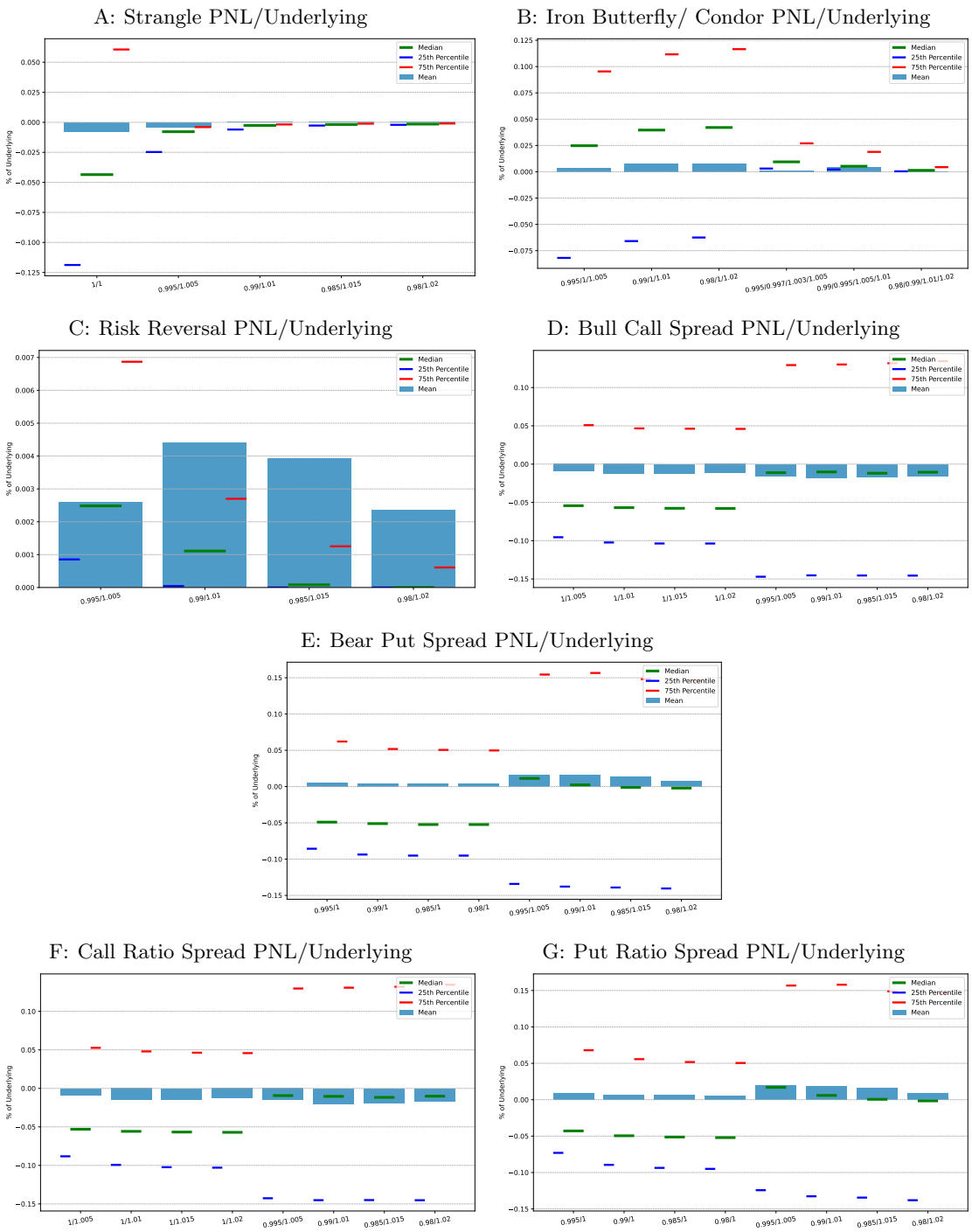


Figure A7: 0DTE Static Option Strategies, 15:00 ET to Expiry. The figure provides statistics on profitability of several well-known option strategies based on 0DTE call and put options; positions are taken at 15:00 ET and held to expiry at 16:00 ET. All panels show option strategy realized profit (payoff - mid-price) per one unit of underlying relative to the underlying price, in %. Bars show mean values, and each bar is accompanied by median, 25th and 75th percentiles. X-axis labels show combination of moneyness for options used for a strategy. The sample period is from 09/2016 to February 2, 2026.

Strategy	Moneyness	Count	Mean	Volatility	Min	1%	25%	50%	75%	99%	Max	Skew	SR, p.a.
Strangle/Straddle	1/1	344	0.0263	0.59	-1.72	-0.96	-0.33	-0.08	0.31	2.10	2.85	1.15	0.71
	0.995/1.005	344	0.0212	0.50	-1.40	-0.79	-0.25	-0.09	0.19	2.01	2.78	1.78	0.67
	0.99/1.01	344	0.0008	0.38	-1.02	-0.49	-0.16	-0.04	-0.01	1.77	2.58	2.90	0.04
	0.985/1.015	344	-0.0029	0.27	-0.73	-0.34	-0.09	-0.02	-0.01	1.43	2.26	4.56	-0.17
	0.98/1.02	344	-0.0049	0.18	-0.50	-0.28	-0.04	-0.01	-0.00	1.05	1.86	6.29	-0.42
Iron Butterfly/Condor	0.995/1/1.005	344	-0.0051	0.16	-0.24	-0.21	-0.11	-0.07	0.11	0.36	0.39	0.89	-0.52
	0.99/1/1.01	344	-0.0254	0.32	-0.71	-0.59	-0.30	-0.05	0.24	0.63	0.70	0.24	-1.26
	0.98/1/1.02	344	-0.0312	0.50	-1.48	-1.19	-0.34	0.04	0.31	0.88	1.22	-0.50	-0.99
	0.995/0.997/1.003/1.005	344	-0.0041	0.09	-0.14	-0.13	-0.07	-0.04	0.08	0.16	0.17	0.46	-0.76
	0.99/0.995/1.005/1.01	344	-0.0203	0.20	-0.47	-0.42	-0.21	0.04	0.13	0.31	0.37	-0.23	-1.64
	0.98/0.99/1.01/1.02	344	-0.0057	0.26	-0.88	-0.79	0.01	0.04	0.13	0.34	0.53	-1.64	-0.35
Risk Reversal	0.995/1.005	344	-0.0001	0.68	-2.84	-1.97	-0.11	0.01	0.09	2.32	3.52	0.24	-0.00
	0.99/1.01	344	0.0105	0.46	-2.35	-1.46	0.00	0.01	0.03	1.83	3.04	0.65	0.36
	0.985/1.015	344	0.0116	0.30	-1.84	-0.95	0.00	0.01	0.02	1.35	2.54	1.56	0.61
	0.98/1.02	344	0.0134	0.19	-1.32	-0.45	0.00	0.00	0.01	0.85	2.02	3.08	1.09
Bull Call Spread	1/1.005	344	0.0000	0.22	-0.26	-0.23	-0.20	-0.13	0.27	0.34	0.37	0.41	0.00
	1/1.01	344	0.0049	0.37	-0.46	-0.41	-0.29	-0.18	0.31	0.75	0.80	0.78	0.21
	1/1.015	344	0.0062	0.45	-0.61	-0.54	-0.32	-0.19	0.26	1.13	1.21	1.09	0.22
	1/1.02	344	0.0063	0.50	-0.71	-0.63	-0.33	-0.19	0.24	1.49	1.59	1.32	0.20
	0.995/1.005	344	-0.0049	0.41	-0.56	-0.55	-0.50	0.04	0.46	0.51	0.51	-0.10	-0.19
	0.99/1.01	344	-0.0161	0.68	-1.12	-1.06	-0.63	0.03	0.55	1.01	1.02	-0.08	-0.38
	0.985/1.015	344	-0.0174	0.82	-1.65	-1.55	-0.63	0.04	0.56	1.50	1.52	-0.08	-0.34
	0.98/1.02	344	-0.0195	0.91	-2.05	-2.02	-0.63	0.05	0.57	1.96	2.03	-0.12	-0.34
Call Ratio Spread	1/1.005	344	-0.0105	0.35	-2.89	-1.55	-0.11	0.02	0.17	0.52	0.59	-3.47	-0.47
	1/1.01	344	-0.0008	0.35	-2.21	-0.89	-0.22	-0.15	0.19	0.74	0.81	-0.38	-0.03
	1/1.015	344	0.0018	0.42	-1.39	-0.44	-0.31	-0.19	0.26	1.13	1.19	1.01	0.07
	1/1.02	344	0.0020	0.47	-0.62	-0.53	-0.33	-0.19	0.25	1.47	1.53	1.31	0.07
	0.995/1.005	344	-0.0155	0.42	-2.69	-1.34	-0.32	-0.01	0.30	0.74	0.81	-1.27	-0.59
	0.99/1.01	344	-0.0218	0.62	-1.88	-1.02	-0.61	0.02	0.47	1.08	1.17	-0.10	-0.56
	0.985/1.015	344	-0.0218	0.78	-1.52	-1.50	-0.62	0.03	0.50	1.48	1.56	-0.13	-0.44
	0.98/1.02	344	-0.0237	0.88	-2.01	-1.97	-0.61	0.05	0.56	1.86	1.95	-0.20	-0.43
Bear Put Spread	0.995/1	344	0.0051	0.22	-0.27	-0.22	-0.18	-0.14	0.29	0.36	0.38	0.56	0.36
	0.99/1	344	0.0205	0.38	-0.40	-0.38	-0.28	-0.17	0.34	0.80	0.86	0.83	0.85
	0.985/1	344	0.0230	0.48	-0.53	-0.50	-0.33	-0.18	0.30	1.17	1.31	1.11	0.77
	0.98/1	344	0.0249	0.54	-0.64	-0.57	-0.34	-0.18	0.29	1.56	1.67	1.36	0.74
	0.995/1.005	344	0.0058	0.41	-0.58	-0.51	-0.46	-0.04	0.50	0.55	0.56	0.10	0.22
	0.99/1.01	324	0.0066	0.68	-1.02	-1.01	-0.58	-0.06	0.63	1.06	1.12	0.10	0.15
	0.985/1.015	298	0.0079	0.85	-1.52	-1.50	-0.62	-0.09	0.64	1.54	1.65	0.13	0.15
	0.98/1.02	272	-0.0066	0.93	-2.03	-1.98	-0.58	-0.10	0.63	2.02	2.05	0.13	-0.11
Put Ratio Spread	0.995/1	344	-0.0055	0.33	-2.22	-1.28	-0.08	0.01	0.16	0.48	0.61	-2.92	-0.26
	0.99/1	344	0.0254	0.34	-1.52	-0.58	-0.19	-0.13	0.22	0.81	0.97	0.44	1.17
	0.985/1	344	0.0302	0.42	-0.70	-0.37	-0.28	-0.17	0.32	1.14	1.26	1.13	1.13
	0.98/1	344	0.0340	0.51	-0.48	-0.47	-0.33	-0.18	0.29	1.54	1.68	1.37	1.06
	0.995/1.005	344	-0.0049	0.39	-2.00	-1.05	-0.25	-0.04	0.30	0.68	0.81	-0.85	-0.20
	0.99/1.01	324	0.0119	0.61	-1.16	-0.98	-0.48	-0.02	0.51	1.19	1.40	0.15	0.31
	0.985/1.015	298	0.0145	0.79	-1.49	-1.46	-0.57	-0.06	0.66	1.56	1.64	0.07	0.29
	0.98/1.02	272	0.0052	0.91	-2.00	-1.91	-0.58	-0.09	0.64	1.99	2.24	0.10	0.09

Table A1: 0DTE Static Option Strategies Performance. The table shows the summary statistics for the holding-period PNL of several well-known option strategies based on 0DTE call and put options; positions are taken at 10:00 ET and held to expiry at 16:00 ET. The statistics are based on option strategies' realized profit per one unit of underlying relative to underlying price $(payoff - mid\ price)/underlying\ price \times 100\%$. The SR, p.a. is the Sharpe Ratio annualized by scaling it up by $\sqrt{252}$. The sample period is from 01/2022 to 12/2023.

Strategy	Moneyness	Count	Mean	Volatility	Min	1%	25%	50%	75%	99%	Max	Skew	SR, p.a.
Strangle/Straddle	1/1	421	-0.0127	0.54	-1.17	-0.90	-0.28	-0.08	0.15	1.70	6.04	4.73	-0.37
	0.995/1.005	421	-0.0106	0.48	-1.25	-0.66	-0.13	-0.07	-0.03	1.64	6.02	6.68	-0.35
	0.99/1.01	421	-0.0026	0.41	-1.10	-0.50	-0.06	-0.02	-0.01	1.47	5.93	9.38	-0.10
	0.985/1.015	421	0.0056	0.36	-1.18	-0.33	-0.02	-0.01	-0.01	1.20	5.78	11.40	0.24
	0.98/1.02	421	0.0100	0.32	-1.11	-0.20	-0.01	-0.00	-0.00	0.85	5.58	13.43	0.49
Iron Butterfly/Condor	0.995/1/1.005	421	0.0021	0.16	-0.28	-0.25	-0.14	-0.04	0.15	0.34	0.40	0.40	0.20
	0.99/1/1.01	421	0.0101	0.28	-0.71	-0.60	-0.19	0.03	0.23	0.56	0.63	-0.30	0.57
	0.98/1/1.02	421	0.0227	0.37	-1.70	-1.26	-0.17	0.07	0.26	0.73	0.87	-1.12	0.97
	0.995/0.997/1.003/1.005	421	0.0021	0.09	-0.16	-0.15	-0.08	0.03	0.07	0.15	0.16	-0.17	0.39
	0.99/0.995/1.005/1.01	421	0.0080	0.16	-0.47	-0.42	-0.01	0.05	0.09	0.28	0.33	-1.17	0.82
	0.98/0.99/1.01/1.02	421	0.0126	0.15	-0.99	-0.86	0.01	0.02	0.05	0.27	0.51	-3.78	1.31
Risk Reversal	0.995/1.005	421	0.0075	0.60	-4.43	-1.75	0.01	0.02	0.04	1.17	8.28	4.68	0.20
	0.99/1.01	421	0.0059	0.49	-3.89	-1.25	0.01	0.01	0.02	0.68	7.80	7.81	0.19
	0.985/1.015	421	0.0072	0.43	-3.36	-0.76	0.00	0.01	0.01	0.19	7.32	10.49	0.27
	0.98/1.02	421	0.0073	0.38	-2.86	-0.27	0.00	0.00	0.01	0.07	6.83	12.92	0.31
Bull Call Spread	1/1.005	421	0.0023	0.20	-0.25	-0.23	-0.17	-0.10	0.21	0.34	0.37	0.51	0.18
	1/1.01	421	-0.0009	0.28	-0.45	-0.40	-0.21	-0.12	0.18	0.72	0.84	0.98	-0.05
	1/1.015	421	-0.0057	0.31	-0.60	-0.49	-0.21	-0.12	0.17	1.00	1.19	1.20	-0.29
	1/1.02	421	-0.0079	0.33	-0.70	-0.54	-0.21	-0.12	0.17	1.08	1.55	1.37	-0.38
	0.995/1.005	421	0.0067	0.36	-0.57	-0.55	-0.30	0.04	0.35	0.50	0.51	-0.17	0.29
	0.99/1.01	421	0.0079	0.52	-1.10	-1.07	-0.29	0.05	0.37	0.99	1.01	-0.24	0.24
	0.985/1.015	421	0.0081	0.59	-1.62	-1.53	-0.30	0.06	0.38	1.43	1.51	-0.35	0.22
	0.98/1.02	421	0.0087	0.62	-2.13	-2.02	-0.29	0.06	0.38	1.63	2.00	-0.47	0.22
Call Ratio Spread	1/1.005	421	0.0038	0.39	-6.89	-0.49	-0.12	-0.05	0.17	0.43	0.75	-13.36	0.16
	1/1.01	421	-0.0025	0.41	-6.33	-0.28	-0.19	-0.12	0.18	0.73	0.81	-8.43	-0.10
	1/1.015	421	-0.0121	0.41	-5.70	-0.37	-0.21	-0.12	0.17	0.97	1.33	-5.64	-0.46
	1/1.02	421	-0.0166	0.40	-5.01	-0.49	-0.22	-0.12	0.17	1.11	1.49	-3.73	-0.65
	0.995/1.005	421	0.0082	0.46	-6.67	-0.52	-0.25	0.07	0.29	0.58	0.84	-7.25	0.28
	0.99/1.01	421	0.0063	0.58	-5.91	-1.05	-0.29	0.06	0.37	1.01	1.11	-2.66	0.17
	0.985/1.015	421	0.0018	0.63	-5.12	-1.51	-0.29	0.06	0.38	1.36	1.94	-1.46	0.04
	0.98/1.02	421	0.0001	0.65	-4.28	-2.00	-0.28	0.06	0.38	1.57	2.22	-1.02	0.00
Bear Put Spread	0.995/1	421	-0.0043	0.20	-0.23	-0.20	-0.15	-0.12	0.15	0.38	0.40	0.89	-0.35
	0.99/1	421	-0.0091	0.30	-0.43	-0.34	-0.20	-0.15	0.11	0.81	0.86	1.37	-0.48
	0.985/1	421	-0.0126	0.36	-0.61	-0.44	-0.22	-0.16	0.09	1.22	1.35	1.73	-0.55
	0.98/1	421	-0.0148	0.39	-0.75	-0.50	-0.22	-0.16	0.09	1.61	1.85	2.07	-0.59
	0.995/1.005	419	-0.0061	0.36	-0.50	-0.50	-0.35	-0.04	0.31	0.55	0.58	0.17	-0.27
	0.99/1.01	410	-0.0111	0.52	-1.02	-0.99	-0.37	-0.05	0.30	1.06	1.11	0.23	-0.34
	0.985/1.015	398	-0.0145	0.59	-1.52	-1.46	-0.38	-0.05	0.29	1.54	1.63	0.33	-0.39
	0.98/1.02	395	-0.0138	0.63	-1.99	-1.68	-0.38	-0.06	0.29	2.03	2.13	0.47	-0.35
Put Ratio Spread	0.995/1	421	0.0047	0.33	-3.54	-1.29	-0.08	-0.04	0.14	0.58	0.91	-5.00	0.23
	0.99/1	421	-0.0049	0.32	-2.85	-0.48	-0.17	-0.13	0.12	0.76	0.91	-1.25	-0.24
	0.985/1	421	-0.0117	0.35	-2.09	-0.33	-0.21	-0.15	0.10	1.11	1.23	1.03	-0.54
	0.98/1	421	-0.0161	0.36	-1.27	-0.39	-0.22	-0.16	0.09	1.19	1.75	1.79	-0.71
	0.995/1.005	419	0.0029	0.40	-3.29	-1.08	-0.26	-0.01	0.31	0.67	0.99	-2.00	0.11
	0.99/1.01	410	-0.0068	0.50	-2.40	-0.96	-0.33	-0.03	0.31	1.02	1.33	-0.08	-0.22
	0.985/1.015	398	-0.0140	0.55	-1.49	-1.39	-0.37	-0.04	0.29	1.45	1.61	0.30	-0.40
	0.98/1.02	395	-0.0155	0.59	-1.94	-1.35	-0.38	-0.06	0.29	1.52	2.09	0.33	-0.42

Table A2: 0DTE Static Option Strategies Performance. The table shows the summary statistics for the holding-period PNL of several well-known option strategies based on 0DTE call and put options; positions are taken at 10:00 ET and held to expiry at 16:00 ET. The statistics are based on option strategies' realized profit per one unit of underlying relative to underlying price $(payoff - mid\ price)/underlying\ price \times 100\%$. The SR, p.a. is the Sharpe Ratio annualized by scaling it up by $\sqrt{252}$. The sample period is from 01/2024 to February 2, 2026.

A.3 Additional Conditional Tables

Symbol	Family	Construction (10:00 ET)	Feature nature
h_i	Liquidity	$\sum_l \lvert q_l \rvert \, bas_l/2$	Half-spread execution-cost proxy at strategy level
d_i	Liquidity	$\sum_l \lvert q_l \rvert (BidSz_l + AskSz_l)$	Displayed depth available at quoted prices
ρ_i	Liquidity	$\sum_l \lvert q_l \rvert \, bas_l / \lvert mid_l \rvert$	Relative spread friction normalized by option price
v_i	Flow	$\sum_l \lvert q_l \rvert \, Vol_l$	Traded volume intensity over strategy legs
f_i^{Δ}	Flow	Leg-sum of $\lvert q_l \rvert Vol_l \lvert \Delta_l \rvert 100 S_l$	Directional traded notional in delta units
f_i^{Γ}	Flow	Leg-sum of $\lvert q_l \rvert Vol_l \lvert \Gamma_l \rvert 100 S_l^2$	Convexity traded notional in gamma dollars
f_i^{ν}	Flow	Leg-sum of $\lvert q_l \rvert Vol_l \lvert \nu_l \rvert 100$	Vol-sensitivity traded notional in vega units
$g_i^{OI,n}$	GEX	Leg-sum of $q_l OI_l \Gamma_l 100 S_l^2$	Signed open-interest-weighted gamma exposure proxy
$g_i^{OI,a}$	GEX	Leg-sum of $\lvert q_l \rvert \lvert OI_l \rvert \lvert \Gamma_l \rvert 100 S_l^2$	Absolute scale of OI-weighted gamma exposure
$g_i^{\Gamma,n}$	GEX	$\sum_l q_l \, \Gamma_l$	Signed structural gamma from strategy-leg mix
$g_i^{\Gamma,a}$	GEX	$\sum_l \lvert q_l \rvert \, \lvert \Gamma_l \rvert$	Absolute structural gamma magnitude
B_i^{Γ}	GEX	$g_i^{OI,n}$ $(\lvert g_i^{OI,a} \rvert + 1)^{-1}$	Normalized sign and balance of OI gamma exposure
R_i^{Γ}	Flow + GEX	f_i^{Γ} $(\lvert g_i^{OI,a} \rvert + 1)^{-1}$	Traded gamma pressure scaled by OI gamma base
T_i	Liquidity	h_i $(d_i + 1)^{-1}$	Cost per displayed depth unit (liquidity tightness)
$M_{i,t}$	Baseline	$\lvert \mathcal{L}_{s,m} \rvert^{-1} \sum_l M_l$	Structure location on the moneyness grid
$P^{(1)}$, $\bar{P}^{(5)}$, $\sigma_P^{(5)}$	Baseline	Lag-1 plus rolling 5-day mean and std within each strategy-moneyness series	Short-horizon persistence and local volatility of strategy PNL
IV_t, IS_t	Market controls	SPXW 0DTE VIX-style level and up-minus-down component at 10:00 ET	Entry-time implied state: overall variance level and asymmetry
Slope and lagged realized controls	Market controls	Slope proxies and lagged realized controls	Recent realized state and smile-shape dynamics (no look-ahead)

Table A3: Conditional Predictor Dictionary for Equations in This Subsection. The table maps the compact notation used in the feature equations to construction formulas and economic interpretation. Family labels are aligned with model-zoo feature groups (Baseline, Flow, GEX, Liquidity); *market controls* are included in all specifications. Duplicate-key preprocessing at $(datetime, option_type, moneyness)$ uses means for price/Greek-like fields (including open interest) and sums for traded/depth flow fields before leg aggregation.

	Strangle	Irons	R/Reversal	C/Spread	C/R/Spread	P/Spread	P/R/Spread
Panel Realized Variance							
RV	0.012	-0.003**	0.009	0.002	-0.008	-0.000	-0.001
	(1.210)	(-2.117)	(0.607)	(0.466)	(-0.839)	(-0.044)	(-0.449)
Combo FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2, Adj.	0.082	0.007	0.037	0.001	0.016	-0.001	0.000
Obs.	5,514	6,607	4,414	8,828	8,828	7,082	7,082
Panel Realized Variance and Skewness							
RV	0.010	-0.004**	-0.007	-0.011**	-0.012*	0.012***	0.004
	(1.222)	(-2.552)	(-0.947)	(-2.103)	(-1.699)	(2.789)	(0.539)
RS	0.014	0.006	0.128***	0.105***	0.033	-0.098***	-0.042*
	(0.411)	(1.239)	(5.131)	(4.275)	(0.899)	(-4.771)	(-1.776)
Combo FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2, Adj.	0.089	0.010	0.606	0.233	0.038	0.253	0.048
Obs.	5,514	6,607	4,414	8,828	8,828	7,082	7,082
Panel Implied and Realized Variances							
IV	-0.042***	0.017***	-0.030	-0.002	0.031**	-0.005	-0.001
	(-2.682)	(4.636)	(-1.157)	(-0.239)	(2.348)	(-0.396)	(-0.143)
RV	0.064**	-0.024***	0.046	0.004	-0.047*	0.006	0.000
	(2.468)	(-4.918)	(1.179)	(0.287)	(-1.902)	(0.342)	(0.017)
Combo FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2, Adj.	0.201	0.041	0.087	-0.000	0.053	-0.000	-0.000
Obs.	5,509	6,601	4,410	8,820	8,820	7,074	7,074
Panel Implied and Realized Variances and Skewness							
IV	-0.038***	0.019***	-0.007	0.012	0.032***	-0.020*	-0.007
	(-3.791)	(4.441)	(-0.871)	(1.116)	(2.904)	(-1.821)	(-0.531)
IS	-0.092***	0.012	-0.086***	0.026	0.114***	-0.034*	-0.032
	(-3.118)	(1.627)	(-4.941)	(1.202)	(3.465)	(-1.699)	(-1.358)
RV	0.056***	-0.027***	-0.000	-0.026	-0.051***	0.037**	0.012
	(5.241)	(-4.550)	(-0.015)	(-1.462)	(-2.952)	(2.046)	(0.622)
RS	-0.016	0.016**	0.110***	0.109***	0.060**	-0.112***	-0.050*
	(-1.051)	(2.514)	(9.195)	(3.942)	(2.342)	(-4.013)	(-1.776)
Combo FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2, Adj.	0.313	0.057	0.693	0.236	0.169	0.286	0.061
Obs.	5,509	6,601	4,410	8,820	8,820	7,074	7,074

Table A4: Option Strategies PNL vs. Implied and Realized Moments. The table shows the results of regressing realized PNL of option strategies (15:00 ET to expiry at 16:00 ET) on implied and realized distribution moments. PNL is specified per one unit of underlying relative to underlying price $(payoff - mid\ price)/underlying\ price \times 100\%$. The result in each column is based on a pooled regression of strategy PNL for several moneyness combinations, including combo fixed effects (Combo FE) and date-clustered standard errors. The sample period is from 09/2016 to February 2, 2026.